

Answers

VARC

1. 3	2. B	3. A	4. D	5. C	6. D	7. A	8. 2143
9. B	10. D	11. A	12. D	13. 1	14. 4312	15. B	16. C
17. D	18. A	19. B	20. D	21. C	22. C	23. A	24. D

LRDI

25. B	26. C	27. C	28. D	29. 41000	30. 13000	31. 1500	32. A
33. A	34. B	35. 2	36. 6	37. D	38. 48	39. 200	40. 1008
41. D	42. C	43. 50	44. 525	45. 350	46. A		

Quant

47. B	48. B	49. 700	50. 65	51. 444	52. A	53. B	54. A
55. D	56. C	57. B	58. 126	59. A	60. B	61. D	62. C
63. A	64. 112	65. C	66. 16	67. 272	68. 397		

Explanations

VARC

1. **3**

The sentences focus on a new scientific discovery: oxygen is being produced on the ocean floor, challenging the common belief that oxygen is produced only through photosynthesis.

We see a link between sentences 5 and 2 - with the structure "don't only" and "they also". These two together introduce the discovery. 4 continues by describing why this discovery is surprising. 1 adds to the discussion in 4 by showing that most oxygen is through photosynthesis making the oxygen production described in 4 an anomaly.

Sentence 3 interrupts this flow because it talks about how the research was done instead of what was discovered, making it the odd one out.

 VIDEO SOLUTION

2. **B**

The passage says people are going back to tailors for three main reasons: they dislike the alike-ness (homogeneity) of fast fashion, are aware of the waste it causes, and want custom clothes that fit well without the high prices of indie or bespoke brands. Option **B** best sums up these points. It shows a move away from both alike-ness and waste in fashion, and also highlights customisation and affordable clothes. These align with the passage's focus on mindfulness, 'fit', sustainability, and affordability without oversimplifying or exaggerating.

Option A only talks about cheaper and better-fitting clothes, missing the points about waste and sameness. Option C is wrong because the passage refers to a mindful group of shoppers, not everyone in India. Option D oversimplifies by focusing just on rejecting expensive brands, while the passage is about leaving fast fashion and avoiding high prices.

VIDEO SOLUTION

3. A

The controversy became “larger” because the 1982 draft act was not seen as an isolated policy mistake, but as a continuation of an older, flawed framework. The passage clearly says that *“the draft act was closely modelled on its predecessor, the Forest Act of 1878.”* This comparison expanded the debate from the immediate provisions of the new act to the entire historical orientation of forest policy. Critics were upset because the 1878 Act was based on *“a usurpation of rights of ownership by the colonial state,”* focused on *“revenue generation and commercial exploitation,”* and left out villagers who had long claimed rights to the forests. Since the 1982 draft act followed this colonial law, it seemed to repeat the same approach of central control and strict rules.

Options (b) and (c) describe why the act was opposed, but they do not explain why the issue widened into a broader historical and structural debate. Option (d) is incomplete, since the larger controversy was not just about commercial exploitation but about how the colonial mechanisms still continued to shape post-Independence forest policy. Hence, option (a) best captures why a raging controversy became a larger one.

VIDEO SOLUTION

4. D

Let us look at each option one by one.

Option (a) would weaken the passage because it shifts responsibility for deforestation onto tribal communities. In the passage, peasants and tribals are presented as groups whose rights were curtailed by forest policy, and not as the primary cause of environmental damage.

Option (b) would weaken the argument too, as it contradicts a key claim made by the author. The passage clearly states that the colonial takeover of forest rights *“had little precedent in precolonial history.”* If Indian rulers had already denied access to forests, this claim would not stand.

Option (c) undermines the historical foundation of the narrative. The passage links the emergence of forest laws to heavy timber demand from railway expansion. If timber was instead imported, the stated reason for creating the forest department would no longer hold.

Option (d) does not weaken the passage because it actually aligns with what the author is already arguing. The passage is critical of colonial forest administration and shows how it sidelined the rights of people living in forest areas. If German forestry experts were known to violate indigenous rights, that would only support this criticism, not challenge it. Hence, option (d) is the correct answer.

VIDEO SOLUTION

5. C

Let's compare each option with what the passage actually states.

Option (a) is common to both. The passage says the 1982 draft act led to a “raging controversy,” and the 1878 Forest Act was passed after a “bitter and prolonged debate within the colonial bureaucracy.” This means both faced debate and opposition.

Option (b) is also shared. The 1982 draft act was criticised for being “closely modelled” on the 1878 Act, which was based on a colonial “usurpation of rights of ownership.” This shows both were viewed as following the same colonial-style control.

Option (c) is not common to both. The passage links large-scale deforestation to the nineteenth century, especially railway expansion before the 1878 Act. In contrast, forest officials said the 1982 draft act aimed to “stop the continuing deforestation,” not cause it.

Option (d) is clearly common to both. The passage criticises both the colonial act and post-Independence policy for keeping “the principle of state monopoly over forest land,” which the government has continued to uphold.

Therefore, option (c) is the feature not shared by both acts.

VIDEO SOLUTION



6. D

The passage indicates that some reforms in forest policy have already taken place. It notes that the government no longer treats forests as a source of revenue and has stopped “ecologically hazardous practices such as the clearfelling of natural forests,” which rules out option (C) since ecological concerns are clearly being recognised. The continued operation of the 1878 Act and the government's reluctance to abandon state control show that the state's claim to forest land remains recognised, ruling out option (A). There is also no mention of a complete ban on deforestation; rather, the focus is on regulating harmful practices, so option (B) does not fit either. What the passage emphasises as still missing is the government's refusal to accept the main demand of critics: “handing over areas of degraded forests to individuals and communities for afforestation.” This directly points to the continued exclusion of local people from forest cultivation and management. Hence, the reform yet to occur is option (D), involving local people in forest cultivation.

VIDEO SOLUTION

7. A

The sentence about the defect being “located to the X chromosome in the region of p11-12” fits best after the description of the abnormal behaviour, i.e., at blank (3). The paragraph starts by introducing the study and its context, then says that “Genetic and metabolic studies were conducted on a large Dutch family...”, and next describes “the undesirable behaviour” like “impulsive aggression, arson and exhibitionism.” After explaining these behaviours, it makes sense to mention the main genetic finding about the defect's location (the X chromosome, region p11-12), and then go into more detail that “a point mutation was identified in the eighth exon of the monoamine oxidase A (MAOA) structural gene...”.

If we put the sentence at blank (1), it would mention the chromosomal location before explaining what kind of study or family was involved. At blank (2), it would break the connection between the study introduction and the explanation that genetic and metabolic studies were done. At blank (4), it would come after the specific point-mutation detail, which should come after the broader chromosomal location. So, the only position that keeps the story moving from study, to methods, to behaviour, to chromosomal location, and then to the precise mutation is blank (3).

VIDEO SOLUTION

8. 2143

This paragraph looks at how photography changes electoral politics. Sentence 2 works best as the opening statement because it introduces the concrete practice: candidates placing their portrait on electoral material, and states that this act “presupposes that photography has a power to convert,” setting up the need for explanation. Sentence 1 should come next because it explains *how* this power works. It shows how the candidate’s image (effigy) creates a personal connection and shifts the focus from political judgment to appearance, posture, and daily habits, which helps explain the point made in sentence 2. Sentence 4 follows by showing the political result. If elections focus on personal image, photography “tends to restore the paternalistic nature of elections,” which goes against party-based or proportional politics. This outcome builds on what sentences 2 and 1 describe. Sentence 3 should be last because it is the most general and evaluative. It sums up the argument by saying photography replaces politics as a set of problems with a “manner of being,” and calls it an anti-intellectual weapon. This makes it a good conclusion, not a starting point.

The ideas should be presented in order: first observation, then analysis, and finally the consequence. So, the best order is 2143.

 VIDEO SOLUTION

9. B

The passage explains that tribal imagination naturally blends different planes of existence. It mentions that tribal stories “*admit fusion between various planes of existence and levels of time in a natural and artless manner*,” and that “*stars, seas, mountains, trees, men and animals can be angry, sad or happy*.” This means emotions are not limited to humans and can pass across boundaries between living forms. What explains non-human forms having human emotions is the acceptance of fluid boundaries between beings, time, and space. Hence, the correct answer is **Option B**.

Option A is not correct because the passage contrasts tribal imagination with secular creativity, which is described as “self-conscious.” Tribal stories are clearly not self-conscious. Option C is also wrong because the author says tribal creations do have conventions and rules. Option D is incorrect because the passage does not call tribal art undeveloped, but instead says it follows a different kind of imagination.

 VIDEO SOLUTION

10. D

The passage explains the difference between imagination and memory by linking them to how humans deal with space and time. It says that we “put meaning into space” by forming images, and this image-making ability is imagination. When it comes to time, we rely on memory to make connections, such as recognising ourselves as the same person across different moments. So imagination helps us understand space, while memory helps us understand time, which leads directly to **option D**.

The other options do not address this main difference. Option A says imagination is a genetic gift, but also claims memory is central to consciousness, which the passage does not mention. Option B talks about tribal art preferences, but the question is about the general difference between imagination and memory. Option C incorrectly says imagination needs to be developed, even though the passage calls it a genetic ability.

 VIDEO SOLUTION

CAT Percentile Predictor



11. A

The passage explains ancestor worship in the context of how tribal communities relate to time rather than space, stating that the tribals realised they could not dominate territorial space, and as a result, they “*turned almost obsessively to gaining domination over time*.” This idea is directly connected to the ritual of conversing with dead ancestors, where tribals aspire to enter a trance and communicate with those from the past. Worship is therefore not just reverence, but a way of overcoming temporal limits. This reasoning clearly leads to option A, which explains ancestor worship as a means of seeking control over time.

Option B only explains the way ancestors are worshipped, not the reason for it. Option C talks about the result of strong memory over time but does not address the main motivation for ancestor worship. Option D is the opposite of what the passage says, since it states that tribals did not try to control land.

 VIDEO SOLUTION

12. D

Let us look at each option one by one.

Option (a) would weaken the passage. The author explains that tribal art can start with “*a trivial everyday event*” and that art blends into daily life. If tribal art left out ordinary reality, the idea of mixing everyday life with imagination would not make sense.

Option (b) would also weaken the argument. The passage points out more than once that there is no strict order, saying “*spatial order and temporal sequence do not restrict the narrative*” and that episodes have “*the chaotic shapes of dreams*.” Having a clear timeline would go against this idea.

Option (c) also directly weakens the passage. The hallucinatory nature of tribal imagination is shown by images that break rational or scientific logic, like oceans flying and mountains swimming. If tribal stories followed scientific rules, this key feature would be lost.

Option (d) does not weaken the passage. The text mentions the ritual of talking with dead ancestors as proof that tribal people engage with time and trance-like states. So, shamanic rituals actually support the idea of a hallucinatory imagination.

So, the correct answer is option (d).

 VIDEO SOLUTION

13. 1

The sentences largely describe music as a universal, complex part of human life. They focus on how music appears in all cultures and is deeply connected to how people think and experience the world.

Sentences 2, 5, 4, and 3 follow a logical order. Sentence 2 starts the paragraph by saying music is universal and involves brain engagement. Sentence 5 logically follows the idea presented in sentence 2 by developing the idea of brain involvement and explaining how cognitively demanding music engagement is. Sentence 4 adds to and broadens this discussion by noting that making and enjoying music are common in all societies and daily life. Sentence 3 connects this idea to human identity and experience and refers to the idea presented in sentence 4.

Sentence 1 does not fit in with the others. Instead of describing what music is or how it works everywhere, it talks about *research inspired by music’s emotional effects*. This brings in a discussion about science, not about music itself. Therefore, the odd sentence out is 1.

VIDEO SOLUTION

14.4312

All four sentences discuss how labels for a community have changed over time, focusing on the shift to gender-neutral terms, the debates about language, and how people actually use these labels. The sequence starts with sentence 4, which introduces the newest change: activists created the term *Latinx* to make the name gender-neutral and more inclusive of LGBTQ people. Sentence 3 comes next, adding that grammatical gender exists in many languages and is different from social or sexual gender. This helps explain why the term "Latinx" was introduced. Sentence 1 then gives a personal response, showing Cherie Moraga's choice to use Latina and Xicana instead of Latinx, which makes the discussion more personal. Sentence 2 is best as the conclusion because it transitions from the individual example to data showing that few people use *Latinx* compared with other terms. So, the best order for the sentences is 4312.

VIDEO SOLUTION

15.B

The paragraph argues that inequality is a feature of modern growth. It explains that because wealth is shaped by assets, institutions, and policy decisions, it grows differently from wages. The paragraph then demonstrates how capital income has been purposefully raised while the mechanisms that once restricted inequality have weakened.

The provided sentence works best at blank 4 because it summarizes the point made immediately before it. The sentence explains the result after claiming that capital income has been "boosted by design" and wages have stagnated: productivity gains now primarily flow to wealth rather than raising living standards. This serves as the argument's obvious conclusion.

Inserting at blank 1 would be premature, as the paragraph hasn't yet explained the distinction between wealth and wages, which the given sentence talks about. Inserting the sentence in blank 2 would interrupt the explanation of how institutions and policy affect wealth. Inserting at blank 3 would interfere with the transition from explanation to consequence because blank 3 comes at a transition point. Therefore, blank 4 is the most appropriate.

VIDEO SOLUTION



16.C

There is a distinct argumentative arc in the passage. Using examples of well-known and pervasive mega-infrastructure, the passage begins by portraying dams as symbols of ambition and extensive control over nature. It then demonstrates how every significant dam serves as a sacrifice zone, uprooting people, devastating ecosystems, and eradicating lifestyles. These losses have been justified as offerings at the "temples of modernity" or in the name of progress. Lastly, the passage notes that global trends continue to favour water impoundment despite mounting evidence of harm and the aging of existing dams.

Option (c), which maps motivation, consequence, justification, and the current trajectory of dam building, best captures this progression. The other options do not represent this overall argument; instead, they list isolated terms or partial effects.

VIDEO SOLUTION

17.D

The Aswan and Hoover dams are referred to in the passage as "charismatic mega-infrastructure," instances that clearly reflect the human desire to dominate land and the natural world. However, the author immediately draws a comparison between these well-known projects and the tens of thousands of smaller dams that collectively alter rivers worldwide but receive far less attention. The goal is to demonstrate that the same mindset that underlies large dams is present at all scales rather than to exalt them. This larger pattern is easier to see thanks to the identifiable symbols of the Hoover and Aswan dams. This leads to option D, which accurately conveys the idea that the desire to control nature is present in both the numerous smaller dams that currently dominate waterways worldwide and the famous mega-dams.

Option A misreads the reference; the "thin blue line" refers to rivers altered by dams, not specifically to the Colorado or Nile. Option B confuses employers and contracts with dams themselves being symbolic structures. Option C wrongly shifts attention to the personal charisma of designers or builders, which the passage never suggests.

VIDEO SOLUTION

18.A

In the first paragraph, dams are called "the physical instantiation of a messianic drive to conquer territories and control nature." Here, *instantiation* means turning an abstract idea into something real. In this case, it is the idea of dominating nature, shown in the form of steel and concrete dams. The dams are more than just buildings; they make that drive real. This meaning is best captured by "exemplification and manifestation," which together convey the sense of something abstract being shown or realised in a tangible form. Hence, option (A) is the correct answer.

The other choices do not fit as well. "Concreteness and viability" (Option B) are about whether something can be done, not about showing an idea. "Durability and timeliness" (Option C) are about how long something lasts and when it happens, which is not the point here. "Development and construction" (Option D) are about building things, but the passage is more about the meaning behind the dams, not how they were built.

VIDEO SOLUTION

19.B

Let us evaluate each option one by one.

Option A can be inferred from the passage. The author notes that "a combination of opposition, alternative energy sources, and a lack of viable projects has slowed the construction of major dams," but immediately adds that "a wave of recent and ongoing construction... continues to tilt the global balance firmly in favor of water impoundment." This clearly supports the idea that dams continue to be built despite resistance and alternatives.

Option B cannot be inferred from the passage. While the author refers to "smaller, though not inconsequential, barriers," this phrase is used to stress their cumulative impact on rivers, not their safety. There is no comparison anywhere in the passage suggesting that smaller dams are safer than large ones. The text discusses ecological damage and displacement caused by dams in general, without ranking them by safety.

Option C is supported by the passage. It explicitly states that "in the western United States, dams were often an instrument of colonialism, used to dispossess Indigenous people and subsidize settler agriculture." This directly implies that dam-building was used to force people off their land.

Option D is also a valid inference. The passage quotes the World Commission on Dams questioning whether we are getting “a fair return from our \$2 trillion investment,” and points out that many dams are ageing, running below capacity, and facing durability issues. Together, this supports the idea that dam-building has been extremely costly and may not be justifiable.

Since option B introduces a claim that the passage does not support, it is the correct answer.

VIDEO SOLUTION

20. D

The passage explains that memory-beliefs exist only in the present and do not require an actual past. It notes that remembering takes place now, and there is no logical reason the remembered event must have happened. The example of the world starting “five minutes ago” shows that our knowledge of the past cannot dismiss this idea. This means memory and knowledge of the past are logically separate from whether the past really happened. Option D addresses the entire argument. It states that knowledge of the past is logically separate from the past itself, links memory-beliefs to what happens in our minds now, and includes the idea that real past events, or even a past at all, are not logically necessary.

The other options do not fully capture the argument. Option A only discusses imagination and misses the logical independence of the past. Option B focuses too much on the five-minute example and leaves out the main point about memory-beliefs. Option C comes close, but it does not clearly state that our knowledge of the past is logically separate from the past itself.

VIDEO SOLUTION



21. C

The question asks which position is most opposite to utilitarianism. According to the passage, utilitarianism is about acting to maximize overall wellbeing or total welfare. The main concern is the total amount of benefit, not who receives it or if some people are worse off, as long as the total outcome is greater. Seen against that, option C moves in the opposite direction. A prioritarian approach does not aim to maximize total welfare. Instead, it gives more importance to helping those who are worst off, even if this does not lead to the highest overall well-being. This challenges the utilitarian focus on total outcomes, making it the closest opposite.

The other options do not match. Option A ranks policies by “overall social welfare,” which is still utilitarian. Option B seems rule-based, but it also talks about “broadest societal benefit,” so it still uses utilitarian thinking. Option D focuses on “highest total benefit,” even though it calls itself deontological, so it is not truly the opposite of utilitarianism. Therefore, the correct answer is option C.

VIDEO SOLUTION

22. C

The passage makes three main points. First, it explains why people find an impersonal AI moral judge appealing. Next, it raises a key concern: moral judgment relies on intuition, history, and context, so turning ethics into rules might remove that depth. The passage also notes that ethics has always included some formal structure, using examples like utilitarianism. It explains with a physics analogy that formal systems can exist together without merging into one final theory. The main idea is not to reject or endorse these systems, but to highlight the tension and variety among them. Option C sums this up best. It shows why an impersonal AI judge is appealing, points out the risk that making rules can “erode case-sensitive judgement,” and still recognizes that reasoning based on principles can be applied widely. By mentioning the physics analogy to explain structured variety, it matches the passage’s idea that ethical systems, like physical theories, can be formal but still different from each other.

Option A goes further than the passage by saying it “rejects formal methods in principle” and that AI should never be used in courts, medicine, or diplomacy “under any conditions.” The passage does not do this; it talks about formal ethical systems and why AI might work for them. Option B misunderstands by saying that codified systems keep case details and that the physics analogy means we will end up with one main framework. In fact, the passage warns about losing nuance and uses the physics example to show that different theories can exist together, not that they will merge. Option D is also not accurate because it treats copying human moral judgment with AI as clear progress. The passage keeps questioning whether copying without depth is really better, so this option is too one-sided.

VIDEO SOLUTION

23. A

The passage says that, like physics, ethics has multiple formal theories, each starting from basic principles, and that AI can reason from such fixed starting points in complex cases. For this comparison to be practically useful, there has to be some way to decide which ethical starting point applies in a given situation. Let us look at each option one by one.

Option A fits the passage. If different ethical frameworks apply to different kinds of cases, then there must be a principled way to decide which framework is relevant before reasoning begins. Otherwise, AI would not know which “starting points” to use. This matches the physics analogy, where different theories apply to different domains.

Option B does not fit. The passage never suggests that real cases fall neatly into one framework with no overlap. In fact, it stresses complexity and divergence across ethical theories, not clean separation.

Option C directly contradicts the passage. The physics analogy is used to show that multiple theories can coexist without being unified into one master framework. This option assumes exactly the opposite.

Option D also conflicts with the text. The passage explicitly notes that ethical theories can diverge and justify similar actions in different ways, so it cannot be assumed that they all produce the same recommendation. Hence, the assumption that must hold is option (A).

VIDEO SOLUTION

24. D

Let us evaluate each option individually.

Option A follows from the passage. The author explains that in physics, different theories are “incompatible” yet share a common structure of starting from basic postulates and reasoning outward. The passage then says ethical theories are similar and also diverge. This supports the idea that ethics can remain plural rather than “close on a single comprehensive framework.”

Option B is also supported. The passage warns that moral judgement draws on “intuition, historical awareness and context” and that formalising ethics risks “flattening its most essential features.” It even says AI could “strip morality of the very depth that makes ethical reflection possible.” This matches the claim that machines would mirror human limits instead of exceeding them.

Option C can reasonably be inferred as well. At the start, AI is described as appealing because it seems “unburdened by emotion, prejudice or inconsistency.” But the passage immediately questions whether this amounts to real moral understanding without “lived experience.” That is exactly the tension this option describes.

Option D does not follow from the passage. The author never predicts convergence on one ethical system. In fact, the physics analogy is used to show that even with formalisation, multiple theories can coexist. The passage also repeatedly insists that context remains central to ethics, not unnecessary.

So the statement that cannot be inferred from the passage is option (D).

[VIDEO SOLUTION](#)

LRDI

25. **B**

Let us place Bina at position 1 and arrange the numbers in a clockwise direction for better understanding, and then position the rest of the people on the table.

Round 1:

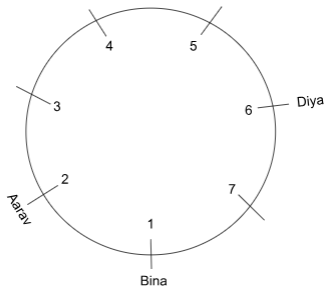
The holder of the buck is Bina. The buck is passed immediately to the left, which means it is given to position 2 from 1, and it is received by Aarav. So, Aarav's position is 2.

Round 2:

The holder of the buck is Aarav. The buck is passed second to the right, which means it is given to the position 7 from 2, and the person receiving it is unknown.

Round 3:

The holder of the buck is unknown, but it is at position 7. The buck is passed immediately to the right, which means it is given to the position 6 from 7, and the person receiving it is Diya. Therefore, Diya's position is 6.



Round 4 and 5:

The buck is at position 6 at the start of round 4 with Diya, and the pass types of rounds 4 and 5 are not provided to us. The only information known to us is that after round 5, the buck is with Aarav, who is at position 2.

The possible ways for Aarav to receive the buck after rounds 4 and 5 are:

- 1) Passing the buck second to the right in round 4 and second to the right in round 5.
- 2) Passing the buck second to the left in round 4 and to the left in round 5.
- 3) Passing the buck to the left in round 4 and second to the left in round 5.

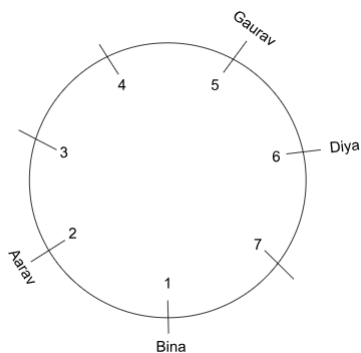
These are the only possible ways for the buck to reach Aarav after round 6.

Round 6:

The holder of the buck is Aarav. The buck is passed second to the left, which means it is given to the position 4 from 2, and the person receiving it is unknown.

Round 7:

The holder of the buck is unknown, but it is at position 4. The buck is passed immediately to the left, which means it is given to position 5 from 4, and the person receiving it is Gaurav. Therefore, Gaurav's position is 5.



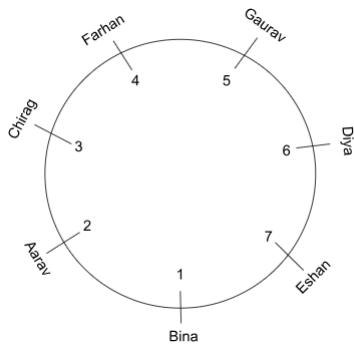
Round 8:

The holder of the buck is Gaurav. The buck is passed immediately to the left, which means it is given to position 6 from 5, and the person receiving it is unknown. But we know that the person in position 6 is Diya. So, the buck is currently at position 6 with Diya.

Round 9 and 10:

The pass types of rounds 9 and 10 are unknown to us, but we are told that the buck is passed to Farhan in round 9 and to Chirag in round 10. The positions of Farhan and Chirag are currently not assigned, and they are out of 3, 4, and 7. From position 6, the only positions that Buck can reach out of the three are 4 or 7. If the buck is passed to position 7 in round 9, then it has to reach either 3 or 4 in round 10 to get to Chirag, but it is not possible to pass the buck from 7 to 3 or 4 in a single round. So, we can eliminate the case of Buck going to 7 in round 9.

The only case left is for the buck to go to position 4 in round 9 and to position 3 in round 10, as from 4, the buck can only be passed to 3 out of 3 and 7 in one round. Hence, we can conclude that Farhan is placed at position 4 and Chirag is placed at position 3. This leaves out only position 7 and the only person left to be assigned is Eshan making the person at position 7 to be Eshan.



This is the final arrangement.

The person immediately to the right of Bina is Eshan.

Hence, the correct answer is option B.

[VIDEO SOLUTION](#)



26. C

Let us place Bina at position 1 and arrange the numbers in a clockwise direction for better understanding, and then position the rest of the people on the table.

Round 1:

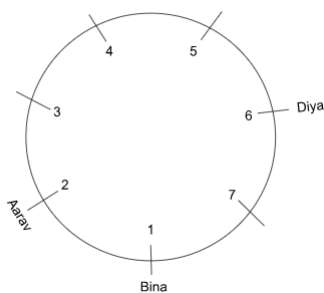
The holder of the buck is Bina. The buck is passed immediately to the left, which means it is given to position 2 from 1, and it is received by Aarav. So, Aarav's position is 2.

Round 2:

The holder of the buck is Aarav. The buck is passed second to the right, which means it is given to the position 7 from 2, and the person receiving it is unknown.

Round 3:

The holder of the buck is unknown, but it is at position 7. The buck is passed immediately to the right, which means it is given to the position 6 from 7, and the person receiving it is Diya. Therefore, Diya's position is 6.



Round 4 and 5:

The buck is at position 6 at the start of round 4 with Diya, and the pass types of rounds 4 and 5 are not provided to us. The only information known to us is that after round 5, the buck is with Aarav, who is at position 2.

The possible ways for Aarav to receive the buck after rounds 4 and 5 are:

- 1) Passing the buck second to the right in round 4 and second to the right in round 5.
- 2) Passing the buck second to the left in round 4 and to the left in round 5.
- 3) Passing the buck to the left in round 4 and second to the left in round 5.

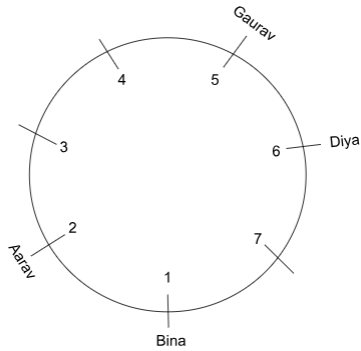
These are the only possible ways for the buck to reach Aarav after round 6.

Round 6:

The holder of the buck is Aarav. The buck is passed second to the left, which means it is given to the position 4 from 2, and the person receiving it is unknown.

Round 7:

The holder of the buck is unknown, but it is at position 4. The buck is passed immediately to the left, which means it is given to position 5 from 4, and the person receiving it is Gaurav. Therefore, Gaurav's position is 5.



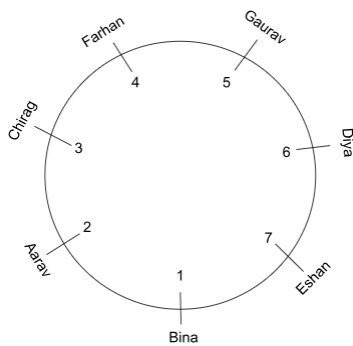
Round 8:

The holder of the buck is Gaurav. The buck is passed immediately to the left, which means it is given to position 6 from 5, and the person receiving it is unknown. But we know that the person in position 6 is Diya. So, the buck is currently at position 6 with Diya.

Round 9 and 10:

The pass types of rounds 9 and 10 are unknown to us, but we are told that the buck is passed to Farhan in round 9 and to Chirag in round 10. The positions of Farhan and Chirag are currently not assigned, and they are out of 3, 4, and 7. From position 6, the only positions that Buck can reach out of the three are 4 or 7. If the buck is passed to position 7 in round 9, then it has to reach either 3 or 4 in round 10 to get to Chirag, but it is not possible to pass the buck from 7 to 3 or 4 in a single round. So, we can eliminate the case of Buck going to 7 in round 9.

The only case left is for the buck to go to position 4 in round 9 and to position 3 in round 10, as from 4, the buck can only be passed to 3 out of 3 and 7 in one round. Hence, we can conclude that Farhan is placed at position 4 and Chirag is placed at position 3. This leaves out only position 7, and the only person left to be assigned is Eshan, making the person at position 7 Eshan.



This is the final arrangement.

The person third to the left of Eshan is Chirag.

Hence, the correct answer is option C.

[VIDEO SOLUTION](#)

27. C

Let us place Bina at position 1 and arrange the numbers in a clockwise direction for better understanding, and then position the rest of the people on the table.

Round 1:

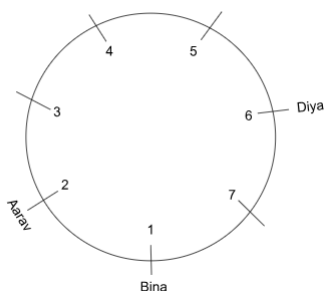
The holder of the buck is Bina. The buck is passed immediately to the left, which means it is given to position 2 from 1, and it is received by Aarav. So, Aarav's position is 2.

Round 2:

The holder of the buck is Aarav. The buck is passed second to the right, which means it is given to the position 7 from 2, and the person receiving it is unknown.

Round 3:

The holder of the buck is unknown, but it is at position 7. The buck is passed immediately to the right, which means it is given to the position 6 from 7, and the person receiving it is Diya. Therefore, Diya's position is 6.



Round 4 and 5:

The buck is at position 6 at the start of round 4 with Diya, and the pass types of rounds 4 and 5 are not provided to us. The only information known to us is that after round 5, the buck is with Aarav, who is at position 2.

The possible ways for Aarav to receive the buck after rounds 4 and 5 are:

- 1) Passing the buck second to the right in round 4 and second to the right in round 5.
- 2) Passing the buck second to the left in round 4 and to the left in round 5.
- 3) Passing the buck to the left in round 4 and second to the left in round 5.

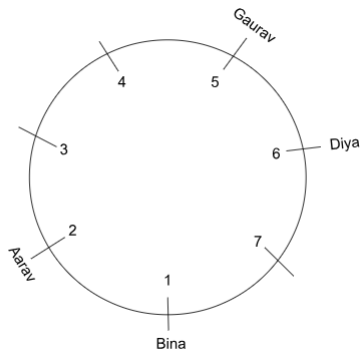
These are the only possible ways for the buck to reach Aarav after round 6.

Round 6:

The holder of the buck is Aarav. The buck is passed second to the left, which means it is given to the position 4 from 2, and the person receiving it is unknown.

Round 7:

The holder of the buck is unknown, but it is at position 4. The buck is passed immediately to the left, which means it is given to position 5 from 4, and the person receiving it is Gaurav. Therefore, Gaurav's position is 5.



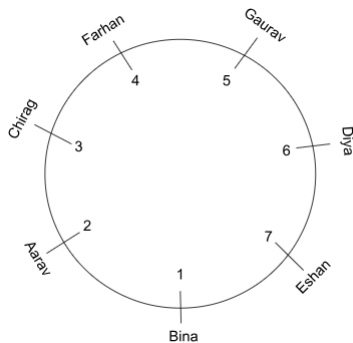
Round 8:

The holder of the buck is Gaurav. The buck is passed immediately to the left, which means it is given to position 6 from 5, and the person receiving it is unknown. But we know that the person in position 6 is Diya. So, the buck is currently at position 6 with Diya.

Round 9 and 10:

The pass types of rounds 9 and 10 are unknown to us, but we are told that the buck is passed to Farhan in round 9 and to Chirag in round 10. The positions of Farhan and Chirag are currently not assigned, and they are out of 3, 4, and 7. From position 6, the only positions that Buck can reach out of the three are 4 or 7. If the buck is passed to position 7 in round 9, then it has to reach either 3 or 4 in round 10 to get to Chirag, but it is not possible to pass the buck from 7 to 3 or 4 in a single round. So, we can eliminate the case of Buck going to 7 in round 9.

The only case left is for the buck to go to position 4 in round 9 and to position 3 in round 10, as from 4, the buck can only be passed to 3 out of 3 and 7 in one round. Hence, we can conclude that Farhan is placed at position 4 and Chirag is placed at position 3. This leaves out only position 7, and the only person left to be assigned is Eshan, making the person at position 7 Eshan.



In rounds 4 and 5, the possibilities were

- 1) Passing the buck second to the right in round 4 and second to the right in round 5.
- 2) Passing the buck second to the left in round 4 and to the left in round 5.
- 3) Passing the buck to the left in round 4 and second to the left in round 5.

Here, Buck could have been passed second to the right, second to the left or immediately to the left in rounds 4 and 5, which is unknown to us. The only pass type that we are certain to have happened is immediately to the right.

Hence, the correct answer is option C.

[VIDEO SOLUTION](#)

28. D

Let us place Bina at position 1 and arrange the numbers in a clockwise direction for better understanding, and then position the rest of the people on the table.

Round 1:

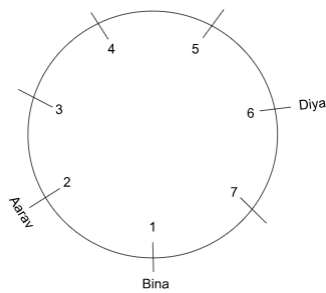
The holder of the buck is Bina. The buck is passed immediately to the left, which means it is given to position 2 from 1, and it is received by Aarav. So, Aarav's position is 2.

Round 2:

The holder of the buck is Aarav. The buck is passed second to the right, which means it is given to the position 7 from 2, and the person receiving it is unknown.

Round 3:

The holder of the buck is unknown, but it is at position 7. The buck is passed immediately to the right, which means it is given to the position 6 from 7, and the person receiving it is Diya. Therefore, Diya's position is 6.



Round 4 and 5:

The buck is at position 6 at the start of round 4 with Diya, and the pass types of rounds 4 and 5 are not provided to us. The only information known to us is that after round 5, the buck is with Aarav, who is at position 2.

The possible ways for Aarav to receive the buck after rounds 4 and 5 are:

- 1) Passing the buck second to the right in round 4 and second to the right in round 5.
- 2) Passing the buck second to the left in round 4 and to the left in round 5.
- 3) Passing the buck to the left in round 4 and second to the left in round 5.

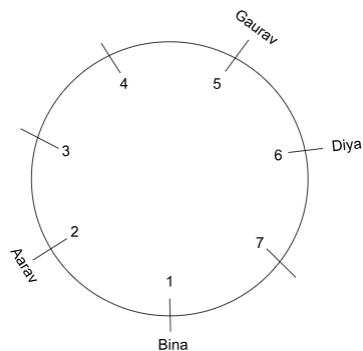
These are the only possible ways for the buck to reach Aarav after round 6.

Round 6:

The holder of the buck is Aarav. The buck is passed second to the left, which means it is given to the position 4 from 2, and the person receiving it is unknown.

Round 7:

The holder of the buck is unknown, but it is at position 4. The buck is passed immediately to the left, which means it is given to position 5 from 4, and the person receiving it is Gaurav. Therefore, Gaurav's position is 5.



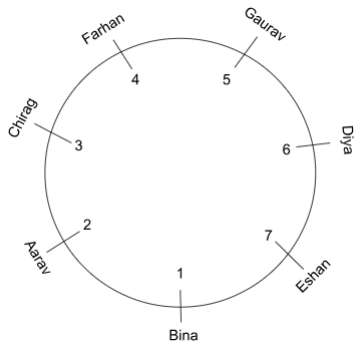
Round 8:

The holder of the buck is Gaurav. The buck is passed immediately to the left, which means it is given to position 6 from 5, and the person receiving it is unknown. But we know that the person in position 6 is Diya. So, the buck is currently at position 6 with Diya.

Round 9 and 10:

The pass types of rounds 9 and 10 are unknown to us, but we are told that the buck is passed to Farhan in round 9 and to Chirag in round 10. The positions of Farhan and Chirag are currently not assigned, and they are out of 3, 4, and 7. From position 6, the only positions that Buck can reach out of the three are 4 or 7. If the buck is passed to position 7 in round 9, then it has to reach either 3 or 4 in round 10 to get to Chirag, but it is not possible to pass the buck from 7 to 3 or 4 in a single round. So, we can eliminate the case of Buck going to 7 in round 9.

The only case left is for the buck to go to position 4 in round 9 and to position 3 in round 10, as from 4, the buck can only be passed to 3 out of 3 and 7 in one round. Hence, we can conclude that Farhan is placed at position 4 and Chirag is placed at position 3. This leaves out only position 7, and the only person left to be assigned is Eshan, making the person at position 7 Eshan.



In rounds 4 and 5, the possibilities were

- 1) Passing the buck second to the right in round 4 and second to the right in round 5. Here, the buck would have gone to Farhan in round 4.
- 2) Passing the buck second to the left in round 4 and to the left in round 5. Here, the buck would have gone to Bina in round 4.
- 3) Passing the buck to the left in round 4 and second to the left in round 5. Here, the buck would have gone to Eshan in round 4.

In round 4, the buck might have gone to either Farhan, Bina or Eshan, which is not known to us. Therefore, the count of the number of times the buck was received by them cannot be determined uniquely, whereas it can be uniquely determined in the case of Gaurav, which is 1 in round 7.

Hence, the correct answer is option D.

[VIDEO SOLUTION](#)

29. 41000

Let us assume units of Aurels, Brins, Crowns, and Zentars to be A, B, C and D, respectively.

We are given the travel cost of Jano to be 3500A in Statement 1.

We are also given the travel cost of Kira to be 8000A in Statement 1 and 4000B in Statement 2.

We are also given the travel cost of Lian to be 36000C in statement 3.

We know that the travel cost has to be constant throughout, so by equating the travel costs of Kira, we get,

$$8000A = 4000B$$

$$B = 2A \text{ ---(1)}$$

We are also given the value of C to be 0.5 Z.

$$C = 0.5Z \text{ ---(2)}$$

$$\text{So, the travel cost of Lian} = 36000C = 36000 \times 0.5Z = 18000Z$$

Let us put the known information in the table, and we get,

TRAVELLER	TRAVEL COST	SPENDING COST		FLIGHT COST
		COUNTRY	MONEY SPENT	
JANO	3500A	Aurevia		4000Z
		Cyrenia		
KIRA	8000A	Aurevia		
		Brelosia	2000B	
LIAN	18000Z	Brelosia	3000B	
		Cyrenia		

We are given that the spending cost of any individual is different in different cities.

We are given that the spending amounts are amongst {1000, 2000, 3000} in their local currencies and also told that there were two spends of 1000 and one spend of 3000, which makes the total spending amount 2000, to be 3 because there are 6 spending amounts in total.

One 2000 and one 3000 are already assigned in the above table, so we will be left with two spends of 1000 and two spends of 2000 to be assigned.

Kira has already spent 2000, so the only amount that she must have spent is 1000, as she cannot spend 2000 twice in different countries.

Lian has already spent 3000, so the only amount that she must have spent is 2000. If she spent 1000, then Jano will be left with two 2000s, which is not possible.

We can also conclude that Jano spend 1000 and 2000 in some order in the two countries.

The flight costs are given as 4000Z, 5000Z, and 6000Z, out of which 4000Z is for Jano and 5000Z and 6000Z are for Kira and Lian, in some order.

Filling up the table with these values, we get,

TRAVELLER	TRAVEL COST	SPENDING COST		FLIGHT COST
		COUNTRY	MONEY SPENT	
JANO	3500A	Aurevia	1000A/2000A	4000Z
		Cyrenia	2000C/1000C	
KIRA	8000A	Aurevia	1000A	5000Z/6000Z
		Brelosia	2000B = 4000A	
LIAN	18000Z	Brelosia	3000B = 6000A	6000Z/5000Z
		Cyrenia	2000C = 1000Z	

Travel Cost = Spending Cost + Flight Cost

For Kira,

$$\text{Travel Cost} = 8000A$$

$$\text{Spending Cost} = 1000A + 4000A = 5000A$$

$$\text{Flight Cost} = 5000Z/6000Z$$

CASE 1: Flight cost of Kira = 5000Z and Flight cost of Lian = 6000Z

If we assume the Flight cost to be 5000Z, then we get,

$$8000A = 5000A + 5000Z$$

$$3000A = 5000Z$$

$$A = \frac{5}{3}Z$$

For Lian,

$$\text{Travel Cost} = 18000Z$$

$$\text{Spending Cost} = 6000A + 1000Z = 6000 \times \frac{5}{3}Z + 1000Z = 11000Z$$

$$\text{Flight Cost} = \text{Travel Cost} - \text{Spending Cost} = 18000Z - 11000Z = 7000Z$$

But in our assumption, the Flight Cost of Lian is 6000Z, which does not match the above answer.

So, we can eliminate this case.

CASE 2: Flight cost of Kira = 6000Z and Flight cost of Lian = 5000Z

If we assume the Flight cost to be 6000Z, then we get,

$$8000A = 5000A + 6000Z$$

$$3000A = 6000Z$$

$$A = 2Z$$

For Lian,

$$\text{Travel Cost} = 18000Z$$

$$\text{Spending Cost} = 6000A + 1000Z = 6000 \cdot 2Z + 1000Z = 13000Z$$

$$\text{Flight Cost} = \text{Travel Cost} - \text{Spending Cost} = 18000Z - 13000Z = 5000Z$$

In our assumption, the Flight Cost of Lian is 5000Z, which matches the above answer.

So, Flight cost of Kira = 6000Z, Flight cost of Lian = 5000Z and A = 2Z.

Filling the table with calculated values all in the currency of Z, we get,

TRAVELLER	TRAVEL COST	SPENDING COST		FLIGHT COST
		COUNTRY	MONEY SPENT	
JANO	7000Z	Aurevia	2000Z/4000Z	4000Z
		Cyrenia	1000Z/500Z	
KIRA	16000Z	Aurevia	2000Z	6000Z
		Brelosia	8000Z	
LIAN	18000Z	Brelosia	12000Z	5000Z
		Cyrenia	1000Z	

In the case of Zano,

$$\text{Spending Cost} = \text{Travel Cost} - \text{Flight Cost} = 7000Z - 4000Z = 3000Z$$

$$\text{In the first case, the Spending Cost} = 2000Z + 1000Z = 3000Z$$

$$\text{In the second case, the Spending Cost} = 4000Z + 500Z = 4500Z$$

We obtained a spending cost of 3000Z only in the first case, so we can eliminate the second case.

The final table looks like,

TRAVELLER	TRAVEL COST	SPENDING COST		FLIGHT COST
		COUNTRY	MONEY SPENT	
JANO	7000Z	Aurevia	2000Z	4000Z
		Cyrenia	1000Z	
KIRA	16000Z	Aurevia	2000Z	6000Z
		Brelosia	8000Z	
LIAN	18000Z	Brelosia	12000Z	5000Z
		Cyrenia	1000Z	

$$\text{The total Travel Cost} = 7000Z + 16000Z + 18000Z = 41000Z$$

Hence, the correct answer is 41000.

 VIDEO SOLUTION

30. 13000

Let us assume units of Aurels, Brins, Crowns, and Zentars to be A, B, C and D, respectively.

We are given the travel cost of Jano to be 3500A in Statement 1.

We are also given the travel cost of Kira to be 8000A in Statement 1 and 4000B in Statement 2.

We are also given the travel cost of Lian to be 36000C in statement 3.

We know that the travel cost has to be constant throughout, so by equating the travel costs of Kira, we get,

$$8000A = 4000B$$

$$B = 2A \text{ ---(1)}$$

We are also given the value of C to be 0.5 Z.

$$C = 0.5Z \text{ ---(2)}$$

$$\text{So, the travel cost of Lian} = 36000C = 36000 \cdot 0.5Z = 18000Z$$

Let us put the known information in the table, and we get,

TRAVELLER	TRAVEL COST	SPENDING COST		FLIGHT COST
		COUNTRY	MONEY SPENT	
JANO	3500A	Aurevia		4000Z
		Cyrenia		
KIRA	8000A	Aurevia		
		Brelosia	2000B	
LIAN	18000Z	Brelosia	3000B	
		Cyrenia		

We are given that the spending cost of any individual is different in different cities.

We are given that the spending amounts are amongst {1000, 2000, 3000} in their local currencies and also told that there were two spends of 1000 and one spend of 3000, which makes the total spending amount 2000, to be 3 because there are 6 spending amounts in total.

One 2000 and one 3000 are already assigned in the above table, so we will be left with two spends of 1000 and two spends of 2000 to be assigned.

Kira has already spent 2000, so the only amount that she must have spent is 1000, as she cannot spend 2000 twice in different countries.

Lian has already spent 3000, so the only amount that she must have spent is 2000. If she spent 1000, then Jano will be left with two 2000s, which is not possible.

We can also conclude that Jano spent 1000 and 2000 in some order in the two countries.

The flight costs are given as 4000Z, 5000Z, and 6000Z, out of which 4000Z is for Jano and 5000Z and 6000Z are for Kira and Lian, in some order.

Filling up the table with these values, we get,

TRAVELLER	TRAVEL COST	SPENDING COST		FLIGHT COST
		COUNTRY	MONEY SPENT	
JANO	3500A	Aurevia	1000A/2000A	4000Z
		Cyrenia	2000C/1000C	
KIRA	8000A	Aurevia	1000A	5000Z/6000Z
		Brelosia	2000B = 4000A	
LIAN	18000Z	Brelosia	3000B = 6000A	6000Z/5000Z
		Cyrenia	2000C = 1000Z	

Travel Cost = Spending Cost + Flight Cost

For Kira,

Travel Cost = 8000A

Spending Cost = 1000A + 4000A = 5000A

Flight Cost = 5000Z/6000Z

CASE 1: Flight cost of Kira = 5000Z and Flight cost of Lian = 6000Z

If we assume the Flight cost to be 5000Z, then we get,

$8000A = 5000A + 5000Z$

$3000A = 5000Z$

$A = \frac{5}{3}Z$

For Lian,

Travel Cost = 18000Z

Spending Cost = $6000A + 1000Z = 6000 \times \frac{5}{3}Z + 1000Z = 11000Z$

Flight Cost = Travel Cost - Spending Cost = $18000Z - 11000Z = 7000Z$

But in our assumption, the Flight Cost of Lian is 6000Z, which does not match the above answer.

So, we can eliminate this case.

CASE 2: Flight cost of Kira = 6000Z and Flight cost of Lian = 5000Z

If we assume the Flight cost to be 6000Z, then we get,

$8000A = 5000A + 6000Z$

$3000A = 6000Z$

$A = 2Z$

For Lian,

Travel Cost = 18000Z

Spending Cost = $6000A + 1000Z = 6000 \times 2Z + 1000Z = 13000Z$

Flight Cost = Travel Cost - Spending Cost = $18000Z - 13000Z = 5000Z$

In our assumption, the Flight Cost of Lian is 5000Z, which matches the above answer.

So, Flight cost of Kira = 6000Z, Flight cost of Lian = 5000Z and $A = 2Z$.

Filling the table with calculated values all in the currency of Z, we get,

TRAVELLER	TRAVEL COST	SPENDING COST		FLIGHT COST
		COUNTRY	MONEY SPENT	
JANO	7000Z	Aurevia	2000Z/4000Z	4000Z
		Cyrenia	1000Z/500Z	
KIRA	16000Z	Aurevia	2000Z	6000Z
		Brelosia	8000Z	
LIAN	18000Z	Brelosia	12000Z	5000Z
		Cyrenia	1000Z	

In the case of Zano,

Spending Cost = Travel Cost - Flight Cost = $7000Z - 4000Z = 3000Z$

In the first case, the Spending Cost = $2000Z + 1000Z = 3000Z$

In the second case, the Spending Cost = $4000Z + 500Z = 4500Z$

We obtained a spending cost of 3000Z only in the first case, so we can eliminate the second case.

The final table looks like,

TRAVELLER	TRAVEL COST	SPENDING COST		FLIGHT COST
		COUNTRY	MONEY SPENT	
JANO	7000Z	Aurevia	2000Z	4000Z
		Cyrenia	1000Z	
KIRA	16000Z	Aurevia	2000Z	6000Z
		Brelosia	8000Z	
LIAN	18000Z	Brelosia	12000Z	5000Z
		Cyrenia	1000Z	

Spending Cost of Lian = $12000Z + 1000Z = 13000Z$

Hence, the correct answer is 13000.



31. 1500

Let us assume units of Aurels, Brins, Crowns, and Zentars to be A, B, C and D, respectively.

We are given the travel cost of Jano to be 3500A in Statement 1.

We are also given the travel cost of Kira to be 8000A in Statement 1 and 4000B in Statement 2.

We are also given the travel cost of Lian to be 36000C in statement 3.

We know that the travel cost has to be constant throughout, so by equating the travel costs of Kira, we get,

$$8000A = 4000B$$

$$B = 2A \text{ ---(1)}$$

We are also given the value of C to be 0.5 Z.

$$C = 0.5Z \text{ ---(2)}$$

$$\text{So, the travel cost of Lian} = 36000C = 36000 \times 0.5Z = 18000Z$$

Let us put the known information in the table, and we get,

TRAVELLER	TRAVEL COST	SPENDING COST		FLIGHT COST
		COUNTRY	MONEY SPENT	
JANO	3500A	Aurevia		4000Z
		Cyrenia		
KIRA	8000A	Aurevia		
		Brelosia	2000B	
LIAN	18000Z	Brelosia	3000B	
		Cyrenia		

We are given that the spending cost of any individual is different in different cities.

We are given that the spending amounts are amongst {1000, 2000, 3000} in their local currencies and also told that there were two spends of 1000 and one spend of 3000, which makes the total spending amount 2000, to be 3 because there are 6 spending amounts in total.

One 2000 and one 3000 are already assigned in the above table, so we will be left with two spends of 1000 and two spends of 2000 to be assigned.

Kira has already spent 2000, so the only amount that she must have spent is 1000, as she cannot spend 2000 twice in different countries.

Lian has already spent 3000, so the only amount that she must have spent is 2000. If she spent 1000, then Jano will be left with two 2000s, which is not possible.

We can also conclude that Jano spend 1000 and 2000 in some order in the two countries.

The flight costs are given as 4000Z, 5000Z, and 6000Z, out of which 4000Z is for Jano and 5000Z and 6000Z are for Kira and Lian, in some order.

Filling up the table with these values, we get,

TRAVELLER	TRAVEL COST	SPENDING COST		FLIGHT COST
		COUNTRY	MONEY SPENT	
JANO	3500A	Aurevia	1000A/2000A	4000Z
		Cyrenia	2000C/1000C	
KIRA	8000A	Aurevia	1000A	5000Z/6000Z
		Brelosia	2000B = 4000A	
LIAN	18000Z	Brelosia	3000B = 6000A	6000Z/5000Z
		Cyrenia	2000C = 1000Z	

Travel Cost = Spending Cost + Flight Cost

For Kira,

$$\text{Travel Cost} = 8000A$$

$$\text{Spending Cost} = 1000A + 4000A = 5000A$$

$$\text{Flight Cost} = 5000Z/6000Z$$

CASE 1: Flight cost of Kira = 5000Z and Flight cost of Lian = 6000Z

If we assume the Flight cost to be 5000Z, then we get,

$$8000A = 5000A + 5000Z$$

$$3000A = 5000Z$$

$$A = \frac{5}{3}Z$$

For Lian,

$$\text{Travel Cost} = 18000Z$$

$$\text{Spending Cost} = 6000A + 1000Z = 6000 \times \frac{5}{3}Z + 1000Z = 11000Z$$

$$\text{Flight Cost} = \text{Travel Cost} - \text{Spending Cost} = 18000Z - 11000Z = 7000Z$$

But in our assumption, the Flight Cost of Lian is 6000Z, which does not match the above answer.

So, we can eliminate this case.

CASE 2: Flight cost of Kira = 6000Z and Flight cost of Lian = 5000Z

If we assume the Flight cost to be 6000Z, then we get,

$$8000A = 5000A + 6000Z$$

$$3000A = 6000Z$$

$$A = 2Z$$

For Lian,

$$\text{Travel Cost} = 18000Z$$

$$\text{Spending Cost} = 6000A + 1000Z = 6000 \cdot 2Z + 1000Z = 13000Z$$

$$\text{Flight Cost} = \text{Travel Cost} - \text{Spending Cost} = 18000Z - 13000Z = 5000Z$$

In our assumption, the Flight Cost of Lian is 5000Z, which matches the above answer.

So, Flight cost of Kira = 6000Z, Flight cost of Lian = 5000Z and $A = 2Z$.

Filling the table with calculated values all in the currency of Z, we get,

TRAVELLER	TRAVEL COST	SPENDING COST		FLIGHT COST
		COUNTRY	MONEY SPENT	
JANO	7000Z	Aurevia	2000Z/4000Z	4000Z
		Cyrenia	1000Z/500Z	
KIRA	16000Z	Aurevia	2000Z	6000Z
		Brelosia	8000Z	
LIAN	18000Z	Brelosia	12000Z	5000Z
		Cyrenia	1000Z	

In the case of Zano,

$$\text{Spending Cost} = \text{Travel Cost} - \text{Flight Cost} = 7000Z - 4000Z = 3000Z$$

$$\text{In the first case, the Spending Cost} = 2000Z + 1000Z = 3000Z$$

$$\text{In the second case, the Spending Cost} = 4000Z + 500Z = 4500Z$$

We obtained a spending cost of 3000Z only in the first case, so we can eliminate the second case.

The final table looks like,

TRAVELLER	TRAVEL COST	SPENDING COST		FLIGHT COST
		COUNTRY	MONEY SPENT	
JANO	7000Z	Aurevia	2000Z	4000Z
		Cyrenia	1000Z	
KIRA	16000Z	Aurevia	2000Z	6000Z
		Brelosia	8000Z	
LIAN	18000Z	Brelosia	12000Z	5000Z
		Cyrenia	1000Z	

$$\text{Total Spending Cost of Jano} = 2000Z + 1000Z = 3000Z$$

We know that $A = 2Z$, so the above value in A can be calculated as,

$$3000Z = 3000 \cdot A / 2 = 1500A$$

So, the spending of Jano in Aurels is 1500A.

Hence, the correct answer is 1500.

[VIDEO SOLUTION](#)

32. A

Let us assume units of Aurels, Brins, Crowns, and Zentars to be A, B, C and D, respectively.

We are given the travel cost of Jano to be 3500A in Statement 1.

We are also given the travel cost of Kira to be 8000A in Statement 1 and 4000B in Statement 2.

We are also given the travel cost of Lian to be 36000C in statement 3.

We know that the travel cost has to be constant throughout, so by equating the travel costs of Kira, we get,

$$8000A = 4000B$$

$$B = 2A \text{ ---(1)}$$

We are also given the value of C to be 0.5 Z.

$$C = 0.5Z \text{ ---(2)}$$

$$\text{So, the travel cost of Lian} = 36000C = 36000 \cdot 0.5Z = 18000Z$$

Let us put the known information in the table, and we get,

TRAVELLER	TRAVEL COST	SPENDING COST		FLIGHT COST
		COUNTRY	MONEY SPENT	
JANO	3500A	Aurevia		4000Z
		Cyrenia		
KIRA	8000A	Aurevia		
		Brelosia	2000B	
LIAN	18000Z	Brelosia	3000B	
		Cyrenia		

We are given that the spending cost of any individual is different in different cities.

We are given that the spending amounts are amongst {1000, 2000, 3000} in their local currencies and also told that there were two spends of 1000 and one spend of 3000, which makes the total spending amount 2000, to be 3 because there are 6 spending amounts in total.

One 2000 and one 3000 are already assigned in the above table, so we will be left with two spends of 1000 and two spends of 2000 to be assigned.

Kira has already spent 2000, so the only amount that she must have spent is 1000, as she cannot spend 2000 twice in different countries.

Lian has already spent 3000, so the only amount that she must have spent is 2000. If she spent 1000, then Jano will be left with two 2000s, which is not possible.

We can also conclude that Jano spent 1000 and 2000 in some order in the two countries.

The flight costs are given as 4000Z, 5000Z, and 6000Z, out of which 4000Z is for Jano and 5000Z and 6000Z are for Kira and Lian, in some order.

Filling up the table with these values, we get,

TRAVELLER	TRAVEL COST	SPENDING COST		FLIGHT COST
		COUNTRY	MONEY SPENT	
JANO	3500A	Aurevia	1000A/2000A	4000Z
		Cyrenia	2000C/1000C	
KIRA	8000A	Aurevia	1000A	5000Z/6000Z
		Brelosia	2000B = 4000A	
LIAN	18000Z	Brelosia	3000B = 6000A	6000Z/5000Z
		Cyrenia	2000C = 1000Z	

Travel Cost = Spending Cost + Flight Cost

For Kira,

Travel Cost = 8000A

Spending Cost = 1000A + 4000A = 5000A

Flight Cost = 5000Z/6000Z

CASE 1: Flight cost of Kira = 5000Z and Flight cost of Lian = 6000Z

If we assume the Flight cost to be 5000Z, then we get,

$8000A = 5000A + 5000Z$

$3000A = 5000Z$

$A = \frac{5}{3}Z$

For Lian,

Travel Cost = 18000Z

Spending Cost = $6000A + 1000Z = 6000 \times \frac{5}{3}Z + 1000Z = 11000Z$

Flight Cost = Travel Cost - Spending Cost = $18000Z - 11000Z = 7000Z$

But in our assumption, the Flight Cost of Lian is 6000Z, which does not match the above answer.

So, we can eliminate this case.

CASE 2: Flight cost of Kira = 6000Z and Flight cost of Lian = 5000Z

If we assume the Flight cost to be 6000Z, then we get,

$8000A = 5000A + 6000Z$

$3000A = 6000Z$

$A = 2Z$

For Lian,

Travel Cost = 18000Z

Spending Cost = $6000A + 1000Z = 6000 \times 2Z + 1000Z = 13000Z$

Flight Cost = Travel Cost - Spending Cost = $18000Z - 13000Z = 5000Z$

In our assumption, the Flight Cost of Lian is 5000Z, which matches the above answer.

So, Flight cost of Kira = 6000Z, Flight cost of Lian = 5000Z and $A = 2Z$.

Filling the table with calculated values all in the currency of Z, we get,

TRAVELLER	TRAVEL COST	SPENDING COST		FLIGHT COST
		COUNTRY	MONEY SPENT	
JANO	7000Z	Aurevia	2000Z/4000Z	4000Z
		Cyrenia	1000Z/500Z	
KIRA	16000Z	Aurevia	2000Z	6000Z
		Brelosia	8000Z	
LIAN	18000Z	Brelosia	12000Z	5000Z
		Cyrenia	1000Z	

In the case of Zano,

Spending Cost = Travel Cost - Flight Cost = $7000Z - 4000Z = 3000Z$

In the first case, the Spending Cost = $2000Z + 1000Z = 3000Z$

In the second case, the Spending Cost = $4000Z + 500Z = 4500Z$

We obtained a spending cost of 3000Z only in the first case, so we can eliminate the second case.

The final table looks like,

TRAVELLER	TRAVEL COST	SPENDING COST		FLIGHT COST
		COUNTRY	MONEY SPENT	
JANO	7000Z	Aurevia	2000Z	4000Z
		Cyrenia	1000Z	
KIRA	16000Z	Aurevia	2000Z	6000Z
		Brelosia	8000Z	
LIAN	18000Z	Brelosia	12000Z	5000Z
		Cyrenia	1000Z	

We know that $B = 2A$ and $A = 2Z$, so the value of B in terms of Z is

$$B = 2 \times 2Z = 4Z \quad \text{---(3)}$$

We also know that

$$C = 0.5Z \quad \text{---(4)}$$

Dividing (3) by (4), we get,

$$\frac{B}{C} = \frac{4Z}{0.5Z} = 8$$

$$B = 8C$$

So, one Brin is equivalent to 8 crowns.

Hence, the correct answer is option A.

[VIDEO SOLUTION](#)

33. A

Let us assume units of Aurels, Brins, Crowns, and Zentars to be A, B, C and D, respectively.

We are given the travel cost of Jano to be 3500A in Statement 1.

We are also given the travel cost of Kira to be 8000A in Statement 1 and 4000B in Statement 2.

We are also given the travel cost of Lian to be 36000C in statement 3.

We know that the travel cost has to be constant throughout, so by equating the travel costs of Kira, we get,

$$8000A = 4000B$$

$$B = 2A \quad \text{---(1)}$$

We are also given the value of C to be 0.5 Z.

$$C = 0.5Z \quad \text{---(2)}$$

$$\text{So, the travel cost of Lian} = 36000C = 36000 \times 0.5Z = 18000Z$$

Let us put the known information in the table, and we get,

TRAVELLER	TRAVEL COST	SPENDING COST		FLIGHT COST
		COUNTRY	MONEY SPENT	
JANO	3500A	Aurevia		4000Z
		Cyrenia		
KIRA	8000A	Aurevia		
		Brelosia	2000B	
LIAN	18000Z	Brelosia	3000B	
		Cyrenia		

We are given that the spending cost of any individual is different in different cities.

We are given that the spending amounts are amongst {1000, 2000, 3000} in their local currencies and also told that there were two spends of 1000 and one spend of 3000, which makes the total spending amount 2000, to be 3 because there are 6 spending amounts in total.

One 2000 and one 3000 are already assigned in the above table, so we will be left with two spends of 1000 and two spends of 2000 to be assigned.

Kira has already spent 2000, so the only amount that she must have spent is 1000, as she cannot spend 2000 twice in different countries.

Lian has already spent 3000, so the only amount that she must have spent is 2000. If she spent 1000, then Jano will be left with two 2000s, which is not possible.

We can also conclude that Jano spend 1000 and 2000 in some order in the two countries.

The flight costs are given as 4000Z, 5000Z, and 6000Z, out of which 4000Z is for Jano and 5000Z and 6000Z are for Kira and Lian, in some order.

Filling up the table with these values, we get,

TRAVELLER	TRAVEL COST	SPENDING COST		FLIGHT COST
		COUNTRY	MONEY SPENT	
JANO	3500A	Aurevia	1000A/2000A	4000Z
		Cyrenia	2000C/1000C	
KIRA	8000A	Aurevia	1000A	5000Z/6000Z
		Brelosia	2000B = 4000A	
LIAN	18000Z	Brelosia	3000B = 6000A	6000Z/5000Z
		Cyrenia	2000C = 1000Z	

Travel Cost = Spending Cost + Flight Cost

For Kira,

$$\text{Travel Cost} = 8000A$$

$$\text{Spending Cost} = 1000A + 4000A = 5000A$$

$$\text{Flight Cost} = 5000Z/6000Z$$

CASE 1: Flight cost of Kira = 5000Z and Flight cost of Lian = 6000Z

If we assume the Flight cost to be 5000Z, then we get,

$$8000A = 5000A + 5000Z$$

$$3000A = 5000Z$$

$$A = \frac{5}{3}Z$$

For Lian,

$$\text{Travel Cost} = 18000Z$$

$$\text{Spending Cost} = 6000A + 1000Z = 6000 \times \frac{5}{3}Z + 1000Z = 11000Z$$

$$\text{Flight Cost} = \text{Travel Cost} - \text{Spending Cost} = 18000Z - 11000Z = 7000Z$$

But in our assumption, the Flight Cost of Lian is 6000Z, which does not match the above answer.

So, we can eliminate this case.

CASE 2: Flight cost of Kira = 6000Z and Flight cost of Lian = 5000Z

If we assume the Flight cost to be 6000Z, then we get,

$$8000A = 5000A + 6000Z$$

$$3000A = 6000Z$$

$$A = 2Z$$

For Lian,

$$\text{Travel Cost} = 18000Z$$

$$\text{Spending Cost} = 6000A + 1000Z = 6000 \times 2Z + 1000Z = 13000Z$$

$$\text{Flight Cost} = \text{Travel Cost} - \text{Spending Cost} = 18000Z - 13000Z = 5000Z$$

In our assumption, the Flight Cost of Lian is 5000Z, which matches the above answer.

So, Flight cost of Kira = 6000Z, Flight cost of Lian = 5000Z and $A = 2Z$.

Filling the table with calculated values all in the currency of Z, we get,

TRAVELLER	TRAVEL COST	SPENDING COST		FLIGHT COST
		COUNTRY	MONEY SPENT	
JANO	7000Z	Aurevia	2000Z/4000Z	4000Z
		Cyrenia	1000Z/500Z	
KIRA	16000Z	Aurevia	2000Z	6000Z
		Brelosia	8000Z	
LIAN	18000Z	Brelosia	12000Z	5000Z
		Cyrenia	1000Z	

In the case of Zano,

$$\text{Spending Cost} = \text{Travel Cost} - \text{Flight Cost} = 7000Z - 4000Z = 3000Z$$

$$\text{In the first case, the Spending Cost} = 2000Z + 1000Z = 3000Z$$

$$\text{In the second case, the Spending Cost} = 4000Z + 500Z = 4500Z$$

We obtained a spending cost of 3000Z only in the first case, so we can eliminate the second case.

The final table looks like,

TRAVELLER	TRAVEL COST	SPENDING COST		FLIGHT COST
		COUNTRY	MONEY SPENT	
JANO	7000Z	Aurevia	2000Z	4000Z
		Cyrenia	1000Z	
KIRA	16000Z	Aurevia	2000Z	6000Z
		Brelosia	8000Z	
LIAN	18000Z	Brelosia	12000Z	5000Z
		Cyrenia	1000Z	

Statement 1) Jano spent 2000 in Aurevia

Jano spent 2000Z in Aurevia = 1000A in Aurevia.

So, this option is incorrect.

Statement 2) Lian spent 2000 in Cyrenia

Lian spent 1000Z in Cyrenia = 2000C in Cyrenia.

So, this option is correct.

Statement 3) Jano spent 2000 in Cyrenia

Jano spent 1000Z in Cyrenia = 2000C in Cyrenia.

So, this option is correct.

Statement 4) Kira spent 1000 in Aurevia

Kira spent 2000Z in Aurevia = 1000A in Aurevia.

So, this option is correct.

Hence, the correct answer is option A.

 VIDEO SOLUTION

34. B

The left plot shows, for each competitor, a step (staircase) curve whose height at any time t equals the number of puzzles that competitor has completed by minute t .

To find who solved the most puzzles by the 20th minute, draw a vertical line at $t=20$ (or simply read the height of each competitor's colored stair at the vertical line $x=20$).

Reading the four curves at $t=20$:

The green staircase (Koushik) is at the highest horizontal level at $t=20$.

The yellow (Chandranath) and red (Anirbid) staircases are lower than the green ones at that time.

The blue (Suranjan) staircase is the lowest of the four at $t=20$.

Since Koushik's value is greater than every other competitor's at $t=20$, Koushik has solved the largest number of puzzles by the 20th minute.

Hence, the correct answer is option B.

[VIDEO SOLUTION](#)

35.2

Comparing the 1st and 2nd graphs, we should be able to determine the type of puzzle.

From the second graph, we can say that there are 4 visual-based puzzles in total, which makes the number-based puzzles 6.

Let us follow the green line in both graphs:

The 1st visual-based puzzle was completed by Green in 2 minutes from the 2nd graph. From the 1st graph at the 2nd minute, the puzzle that was solved by Green was 1. So, we can say that 1 is a visual-based puzzle.

The 2nd visual-based puzzle was completed by Green in 12 minutes from the 2nd graph. From the 1st graph at the 12th minute, the puzzle that was solved by Green was 4. So, we can say that 4 is a visual-based puzzle.

The 3rd visual-based puzzle was completed by Green in 25 minutes from the 2nd graph. From the 1st graph at the 25th minute, the puzzle that was solved by Green was 8. So, we can say that 8 is a visual-based puzzle.

The 4th visual-based puzzle was completed by Green in 29 minutes from the 2nd graph. From the 1st graph at the 29th minute, the puzzle that was solved by Green was 9. So, we can say that 9 is a visual-based puzzle.

We can conclude that puzzles 1, 4, 8, and 9 are visual-based, while puzzles 2, 3, 5, 6, 7, and 10 are number-based.

So, the third visual puzzle is 8, and the time taken by Suranjan(blue) to complete the 3rd visual-based puzzle, which is the 8th puzzle, can be calculated as,

The time taken by Suranjan to solve the 8th puzzle from graph 1 = $28 - 26 = 2$ minutes.

Hence, the correct answer is 2.

[VIDEO SOLUTION](#)



36.6

Comparing the 1st and 2nd graphs, we should be able to determine the type of puzzle.

From the second graph, we can say that there are 4 visual-based puzzles in total, which makes the number-based puzzles 6.

Let us follow the green line in both graphs:

The 1st visual-based puzzle was completed by Green in 2 minutes from the 2nd graph. From the 1st graph at the 2nd minute, the puzzle that was solved by Green was 1. So, we can say that 1 is a visual-based puzzle.

The 2nd visual-based puzzle was completed by Green in 12 minutes from the 2nd graph. From the 1st graph at the 12th minute, the puzzle that was solved by Green was 4. So, we can say that 4 is a visual-based puzzle.

The 3rd visual-based puzzle was completed by Green in 25 minutes from the 2nd graph. From the 1st graph at the 25th minute, the puzzle that was solved by Green was 8. So, we can say that 8 is a visual-based puzzle.

The 4th visual-based puzzle was completed by Green in 29 minutes from the 2nd graph. From the 1st graph at the 29th minute, the puzzle that was solved by Green was 9. So, we can say that 9 is a visual-based puzzle.

We can conclude that puzzles 1, 4, 8, and 9 are visual-based, while puzzles 2, 3, 5, 6, 7, and 10 are number-based.

The fourth number-based puzzle is 6th puzzle in the order.

Hence, the correct answer is 6.

[VIDEO SOLUTION](#)

37.D

Comparing the 1st and 2nd graphs, we should be able to determine the type of puzzle.

From the second graph, we can say that there are 4 visual-based puzzles in total, which makes the number-based puzzles 6.

Let us follow the green line in both graphs:

The 1st visual-based puzzle was completed by Green in 2 minutes from the 2nd graph. From the 1st graph at the 2nd minute, the puzzle that was solved by Green was 1. So, we can say that 1 is a visual-based puzzle.

The 2nd visual-based puzzle was completed by Green in 12 minutes from the 2nd graph. From the 1st graph at the 12th minute, the puzzle that was solved by Green was 4. So, we can say that 4 is a visual-based puzzle.

The 3rd visual-based puzzle was completed by Green in 25 minutes from the 2nd graph. From the 1st graph at the 25th minute, the puzzle that was solved by Green was 8. So, we can say that 8 is a visual-based puzzle.

The 4th visual-based puzzle was completed by Green in 29 minutes from the 2nd graph. From the 1st graph at the 29th minute, the puzzle that was solved by Green was 9. So, we can say that 9 is a visual-based puzzle.

We can conclude that puzzles 1, 4, 8, and 9 are visual-based, while puzzles 2, 3, 5, 6, 7, and 10 are number-based.

The average time taken by Anirbid to solve the number-based puzzles can be calculated as,

Time for 1st number-based puzzle(2nd puzzle) from 1st graph = $9 - 4 = 5$ minutes.

Time for 2nd number-based puzzle(3rd puzzle) from 1st graph = $11 - 9 = 2$ minutes.

Time for 3rd number-based puzzle(5th puzzle) from 1st graph = $19 - 14 = 5$ minutes.

Time for 4th number-based puzzle(6th puzzle) from 1st graph = $22 - 19 = 3$ minutes.

Time for 5th number-based puzzle(7th puzzle) from 1st graph = $27 - 22 = 5$ minutes.

Time for 6th number-based puzzle(10th puzzle) from 1st graph = $37 - 33 = 4$ minutes.

$$\text{Average} = \frac{5 + 2 + 5 + 3 + 5 + 4}{6} = \frac{24}{6} = 4 \text{ minutes}$$

The average time taken by Anirbid to solve the number-based puzzles is 4 minutes.

Hence, the correct answer is option D.

[VIDEO SOLUTION](#)

38.48

In clue 5, we are given that export volumes of P, to X and C are 600 and 1200, respectively, and P is the only country that exports to C. So, the export from X to C and ROW to C is 0.

In clue 1, we are given that normalised trade balances of P, X and C are 0%, 10%, and -20%, respectively.

For P, since the normalised trade balance is 0% the exports have to be equal to the imports according to the given formula. So, let us assume them to be 10a and 10a.

For X,

$$\frac{E - I}{E + I} \times 100 = 10$$

$$10E - 10I = E + I$$

$$11I = 9E$$

So, let us assume export to be 11b and import to be 9b.

For C,

$$\frac{E - I}{E + I} \times 100 = -20$$

$$5I - 5E = E + I$$

$$2I = 3E$$

So, let us assume export to be 2c and import to be 3c.

	P		X		C		ROW		TOTAL	
	IMPORTS	EXPORTS	IMPORTS	EXPORTS	IMPORTS	EXPORTS	IMPORTS	EXPORTS	IMPORTS	EXPORTS
P	0	0		600		1200			10a	10a
X	600		0	0					9b	11b
C	1200				0	0			3c	2c
ROW										

In clue 2, we are given that 40% of exports of X are to P and 22% of imports of P are from X.

$$40\% \text{ of the exports of X} = \frac{40}{100} \times 11b = 4.4b = \text{exports from X to P}$$

$$22\% \text{ of the imports of P} = \frac{22}{100} \times 10a = 2.2a = \text{imports to P from X}$$

Equating both, we get,

$$4.4b = 2.2a$$

$$a = 2b.$$

We are also given that imports for C are only from P, and we can set the rest of the imports for C to 0.

The total imports for C = $3c = 1200$

$$c = 400$$

This makes the exports for C = $2c = 800$

In clue 3, we are given that 90% of exports of C are to P and 4% are to ROW.

$$90\% \text{ of the exports of C} = \frac{90}{100} \times 2c = 1.8c = 1.8 \times 400 = 720 = \text{exports from C to P}$$

$$4\% \text{ of the exports of C} = \frac{4}{100} \times 2c = 0.08c = 0.08 \times 400 = 32 = \text{exports from C to ROW}$$

$$\text{The exports from C to X} = 800 - 720 - 32 = 48$$

In clue 4, we are given that 12% of exports of ROW are to X, 40% are to P.

If we assume that Imports of ROW are m and exports of ROW are n.

$$12\% \text{ of the exports of ROW} = \frac{12}{100} \times n = 0.12n = \text{exports from ROW to X}$$

$$40\% \text{ of the exports of ROW} = \frac{40}{100} \times n = 0.4n = \text{exports from ROW to P}$$

Placing all the values in the table, we get,

	P		X		C		ROW		TOTAL	
	IMPORTS	EXPORTS	IMPORTS	EXPORTS	IMPORTS	EXPORTS	IMPORTS	EXPORTS	IMPORTS	EXPORTS
P	0	0	4.4b	600	720	1200	0.4n		20b	20b
X	600	4.4b	0	0	48	0	0.12n		9b	11b
C	1200	720	0	48	0	0	0	32	1200	800
ROW		0.4n		0.12n	32	0				

Equating the imports of P and X to the Totals, we get,

For P,

$$4.4b + 720 + 0.4n = 20b$$

$$15.6b = 720 + 0.4n \quad \text{--(1)}$$

For X,

$$600 + 48 + 0.12n = 9b$$

$$9b = 648 + 0.12n \quad \text{--(2)}$$

Solving (1) and (2), we get

$$\frac{648 + 0.12n}{9} = \frac{720 + 0.4n}{15.6}$$

$$10108.8 + 1.872n = 6480 + 3.6n$$

$$3628.8 = 1.728n$$

$$n = 2100$$

From (1),

$$15.6b = 720 + 0.4(2100)$$

$$15.6b = 1560$$

$$b = 100$$

$$\text{Exports of P to ROW} = 20b - 600 - 1200 = 2000 - 1800 = 200$$

$$\text{Exports of X to ROW} = 11b - 4.4b = 6.6b = 660$$

Substituting the values of b and n, we can figure out all the values except for Imports and Exports of ROW to ROW.

The value of exports and imports of ROW to ROW has to be equal.

We calculated the value of n to be 2100.

The exports of ROW to ROW can be calculated as,

$$\text{Exports of ROW to ROW} = 2100 - 840 - 252 = 1008 = \text{Imports of ROW to ROW.}$$

The value of m can be calculated as,

$$m = 200 + 660 + 32 + 1008 = 1900.$$

Filling up the table with all the values, we get,

	P		X		C		ROW		TOTAL	
	IMPORTS	EXPORTS	IMPORTS	EXPORTS	IMPORTS	EXPORTS	IMPORTS	EXPORTS	IMPORTS	EXPORTS
P	0	0	440	600	720	1200	840	200	2000	2000
X	600	440	0	0	48	0	252	660	900	1100
C	1200	720	0	48	0	0	0	32	1200	800
ROW	200	840	660	252	32	0	1008	1008	1900	2100

Exports from C to X are 48 IC.

Hence, the correct answer is 48.

VIDEO SOLUTION

39.200

In clue 5, we are given that export volumes of P, in IC, to X and C are 600 and 1200, respectively, and P is the only country that exports to C. So, the export from X to C and ROW to C is 0.

In clue 1, we are given that normalised trade balances of P, X and C are 0%, 10%, and -20%, respectively.

For P, since the normalised trade balance is 0% the exports have to be equal to the imports according to the given formula. So, let us assume them to be 10a and 10a.

For X,

$$\frac{E - I}{E + I} \times 100 = 10$$

$$10E - 10I = E + I$$

$$11I = 9E$$

So, let us assume export to be 11b and import to be 9b.

For C,

$$\frac{E - I}{E + I} \times 100 = -20$$

$$5I - 5E = E + I$$

$$2I = 3E$$

So, let us assume export to be 2c and import to be 3c.

	P		X		C		ROW		TOTAL	
	IMPORTS	EXPORTS	IMPORTS	EXPORTS	IMPORTS	EXPORTS	IMPORTS	EXPORTS	IMPORTS	EXPORTS
P	0	0		600		1200			10a	10a
X	600		0	0					9b	11b
C	1200				0	0			3c	2c
ROW										

In clue 2, we are given that 40% of exports of X are to P and 22% of imports of P are from X.

$$40\% \text{ of the exports of X} = \frac{40}{100} \times 11b = 4.4b = \text{exports from X to P}$$

$$22\% \text{ of the imports of P} = \frac{22}{100} \times 10a = 2.2a = \text{imports to P from X}$$

Equating both, we get,

$$4.4b = 2.2a$$

$$a = 2b.$$

We are also given that imports for C are only from P, and we can set the rest of the imports for C to 0.

The total imports for C = 3c = 1200

$$c = 400$$

This makes the exports for C = 2c = 800

In clue 3, we are given that 90% of exports of C are to P and 4% are to ROW.

$$90\% \text{ of the exports of C} = \frac{90}{100} \times 2c = 1.8c = 1.8 \times 400 = 720 = \text{exports from C to P}$$

$$4\% \text{ of the exports of C} = \frac{4}{100} \times 2c = 0.08c = 0.08 \times 400 = 32 = \text{exports from C to ROW}$$

$$\text{The exports from C to X} = 800 - 720 - 32 = 48$$

In clue 4, we are given that 12% of exports of ROW are to X, 40% are to P.

If we assume that Imports of ROW are m and exports of ROW are n.

$$12\% \text{ of the exports of ROW} = \frac{12}{100} \times n = 0.12n = \text{exports from ROW to X}$$

$$40\% \text{ of the exports of ROW} = \frac{40}{100} \times n = 0.4n = \text{exports from ROW to P}$$

Placing all the values in the table, we get,

	P		X		C		ROW		TOTAL	
	IMPORTS	EXPORTS	IMPORTS	EXPORTS	IMPORTS	EXPORTS	IMPORTS	EXPORTS	IMPORTS	EXPORTS
P	0	0	4.4b	600	720	1200	0.4n		20b	20b
X	600	4.4b	0	0	48	0	0.12n		9b	11b
C	1200	720	0	48	0	0	0	32	1200	800
ROW		0.4n		0.12n	32	0				

Equating the imports of P and X to the Totals, we get,

For P,

$$4.4b + 720 + 0.4n = 20b$$

$$15.6b = 720 + 0.4n \quad (1)$$

For X,

$$600 + 48 + 0.12n = 9b$$

$$9b = 648 + 0.12n \quad (2)$$

Solving (1) and (2), we get

$$\frac{648 + 0.12n}{9} = \frac{720 + 0.4n}{15.6}$$

$$10108.8 + 1.872n = 6480 + 3.6n$$

$$3628.8 = 1.728n$$

$$n = 2100$$

From (1),

$$15.6b = 720 + 0.4(2100)$$

$$15.6b = 1560$$

$$b = 100$$

$$\text{Exports of P to ROW} = 20b - 600 - 1200 = 2000 - 1800 = 200$$

$$\text{Exports of X to ROW} = 11b - 4.4b = 6.6b = 660$$

Substituting the values of b and n, we can figure out all the values except for Imports and Exports of ROW to ROW.

The value of exports and imports of ROW to ROW has to be equal.

We calculated the value of n to be 2100.

The exports of ROW to ROW can be calculated as,

$$\text{Exports of ROW to ROW} = 2100 - 840 - 252 = 1008 = \text{Imports of ROW to ROW.}$$

The value of m can be calculated as,

$$m = 200 + 660 + 32 + 1008 = 1900.$$

Filling up the table with all the values, we get,

	P		X		C		ROW		TOTAL	
	IMPORTS	EXPORTS	IMPORTS	EXPORTS	IMPORTS	EXPORTS	IMPORTS	EXPORTS	IMPORTS	EXPORTS
P	0	0	440	600	720	1200	840	200	2000	2000
X	600	440	0	0	48	0	252	660	900	1100
C	1200	720	0	48	0	0	0	32	1200	800
ROW	200	840	660	252	32	0	1008	1008	1900	2100

The exports from P to ROW 200 IC.

Hence, the correct answer is 200.

[VIDEO SOLUTION](#)

40. 1008

In clue 5, we are given that export volumes of P, in IC, to X and C are 600 and 1200, respectively, and P is the only country that exports to C. So, the export from X to C and ROW to C is 0.

In clue 1, we are given that normalised trade balances of P, X and C are 0%, 10%, and -20%, respectively.

For P, since the normalised trade balance is 0% the exports have to be equal to the imports according to the given formula. So, let us assume them to be 10a and 10a.

For X,

$$\frac{E - I}{E + I} \times 100 = 10$$

$$10E - 10I = E + I$$

$$11I = 9E$$

So, let us assume export to be 11b and import to be 9b.

For C,

$$\frac{E - I}{E + I} \times 100 = -20$$

$$5I - 5E = E + I$$

$$2I = 3E$$

So, let us assume export to be 2c and import to be 3c.

	P		X		C		ROW		TOTAL	
	IMPORTS	EXPORTS	IMPORTS	EXPORTS	IMPORTS	EXPORTS	IMPORTS	EXPORTS	IMPORTS	EXPORTS
P	0	0		600		1200			10a	10a
X	600		0	0					9b	11b
C	1200				0	0			3c	2c
ROW										

In clue 2, we are given that 40% of exports of X are to P and 22% of imports of P are from X.

$$40\% \text{ of the exports of X} = \frac{40}{100} \times 11b = 4.4b = \text{exports from X to P}$$

$$22\% \text{ of the imports of P} = \frac{22}{100} \times 10a = 2.2a = \text{imports to P from X}$$

Equating both, we get,

$$4.4b = 2.2a$$

$$a = 2b.$$

We are also given that imports for C are only from P, and we can set the rest of the imports for C to 0.

The total imports for C = 3c = 1200

$$c = 400$$

This makes the exports for C = $2c = 800$

In clue 3, we are given that 90% of exports of C are to P and 4% are to ROW.

$$90\% \text{ of the exports of C} = \frac{90}{100} \times 2c = 1.8c = 1.8 \times 400 = 720 = \text{exports from C to P}$$

$$4\% \text{ of the exports of C} = \frac{4}{100} \times 2c = 0.08c = 0.08 \times 400 = 32 = \text{exports from C to ROW}$$

$$\text{The exports from C to X} = 800 - 720 - 32 = 48$$

In clue 4, we are given that 12% of exports of ROW are to X, 40% are to P.

If we assume that Imports of ROW are m and exports of ROW are n .

$$12\% \text{ of the exports of ROW} = \frac{12}{100} \times n = 0.12n = \text{exports from ROW to X}$$

$$40\% \text{ of the exports of ROW} = \frac{40}{100} \times n = 0.4n = \text{exports from ROW to P}$$

Placing all the values in the table, we get,

	P		X		C		ROW		TOTAL	
	IMPORTS	EXPORTS	IMPORTS	EXPORTS	IMPORTS	EXPORTS	IMPORTS	EXPORTS	IMPORTS	EXPORTS
P	0	0	4.4b	600	720	1200	0.4n		20b	20b
X	600	4.4b	0	0	48	0	0.12n		9b	11b
C	1200	720	0	48	0	0	0	32	1200	800
ROW		0.4n		0.12n	32	0				

Equating the imports of P and X to the Totals, we get,

For P,

$$4.4b + 720 + 0.4n = 20b$$

$$15.6b = 720 + 0.4n \quad \text{---(1)}$$

For X,

$$600 + 48 + 0.12n = 9b$$

$$9b = 648 + 0.12n \quad \text{---(2)}$$

Solving (1) and (2), we get

$$\frac{648 + 0.12n}{9} = \frac{720 + 0.4n}{15.6}$$

$$10108.8 + 1.872n = 6480 + 3.6n$$

$$3628.8 = 1.728n$$

$$n = 2100$$

From (1),

$$15.6b = 720 + 0.4(2100)$$

$$15.6b = 1560$$

$$b = 100$$

$$\text{Exports of P to ROW} = 20b - 600 - 1200 = 2000 - 1800 = 200$$

$$\text{Exports of X to ROW} = 11b - 4.4b = 6.6b = 660$$

Substituting the values of b and n , we can figure out all the values except for Imports and Exports of ROW to ROW.

The value of exports and imports of ROW to ROW has to be equal.

We calculated the value of n to be 2100.

The exports of ROW to ROW can be calculated as,

$$\text{Exports of ROW to ROW} = 2100 - 840 - 252 = 1008 = \text{Imports of ROW to ROW.}$$

The value of m can be calculated as,

$$m = 200 + 660 + 32 + 1008 = 1900.$$

Filling up the table with all the values, we get,

	P		X		C		ROW		TOTAL	
	IMPORTS	EXPORTS	IMPORTS	EXPORTS	IMPORTS	EXPORTS	IMPORTS	EXPORTS	IMPORTS	EXPORTS
P	0	0	440	600	720	1200	840	200	2000	2000
X	600	440	0	0	48	0	252	660	900	1100
C	1200	720	0	48	0	0	0	32	1200	800
ROW	200	840	660	252	32	0	1008	1008	1900	2100

The exports from ROW to ROW are 1008 IC.

Hence, the correct answer is 1008.

VIDEO SOLUTION

41. D

In clue 5, we are given that export volumes of P, in IC, to X and C are 600 and 1200, respectively, and P is the only country that exports to C. So, the export from X to C and ROW to C is 0.

In clue 1, we are given that normalised trade balances of P, X and C are 0%, 10%, and -20%, respectively.

For P, since the normalised trade balance is 0% the exports have to be equal to the imports according to the given formula. So, let us assume them to be 10a and 10a.

For X,

$$\frac{E - I}{E + I} \times 100 = 10$$

$$10E - 10I = E + I$$

$$11I = 9E$$

So, let us assume export to be 11b and import to be 9b.

For C,

$$\frac{E - I}{E + I} \times 100 = -20$$

$$5I - 5E = E + I$$

$$2I = 3E$$

So, let us assume export to be 2c and import to be 3c.

	P		X		C		ROW		TOTAL	
	IMPORTS	EXPORTS	IMPORTS	EXPORTS	IMPORTS	EXPORTS	IMPORTS	EXPORTS	IMPORTS	EXPORTS
P	0	0		600		1200			10a	10a
X	600		0	0					9b	11b
C	1200				0	0			3c	2c
ROW										

In clue 2, we are given that 40% of exports of X are to P and 22% of imports of P are from X.

$$40\% \text{ of the exports of X} = \frac{40}{100} \times 11b = 4.4b = \text{exports from X to P}$$

$$22\% \text{ of the imports of P} = \frac{22}{100} \times 10a = 2.2a = \text{imports to P from X}$$

Equating both, we get,

$$4.4b = 2.2a$$

$$a = 2b.$$

We are also given that imports for C are only from P, and we can set the rest of the imports for C to 0.

The total imports for C = 3c = 1200

$$c = 400$$

This makes the exports for C = 2c = 800

In clue 3, we are given that 90% of exports of C are to P and 4% are to ROW.

$$90\% \text{ of the exports of C} = \frac{90}{100} \times 2c = 1.8c = 1.8 \times 400 = 720 = \text{exports from C to P}$$

$$4\% \text{ of the exports of C} = \frac{4}{100} \times 2c = 0.08c = 0.08 \times 400 = 32 = \text{exports from C to ROW}$$

The exports from C to X = 800 - 720 - 32 = 48

In clue 4, we are given that 12% of exports of ROW are to X, 40% are to P.

If we assume that Imports of ROW are m and exports of ROW are n.

$$12\% \text{ of the exports of ROW} = \frac{12}{100} \times n = 0.12n = \text{exports from ROW to X}$$

$$40\% \text{ of the exports of ROW} = \frac{40}{100} \times n = 0.4n = \text{exports from ROW to P}$$

Placing all the values in the table, we get,

	P		X		C		ROW		TOTAL	
	IMPORTS	EXPORTS	IMPORTS	EXPORTS	IMPORTS	EXPORTS	IMPORTS	EXPORTS	IMPORTS	EXPORTS
P	0	0	4.4b	600	720	1200	0.4n		20b	20b
X	600	4.4b	0	0	48	0	0.12n		9b	11b
C	1200	720	0	48	0	0	0	32	1200	800
ROW		0.4n		0.12n	32	0				

Equating the imports of P and X to the Totals, we get,

For P,

$$4.4b + 720 + 0.4n = 20b$$

$$15.6b = 720 + 0.4n \quad (1)$$

For X,

$$600 + 48 + 0.12n = 9b$$

$$9b = 648 + 0.12n \quad (2)$$

Solving (1) and (2), we get

$$\frac{648 + 0.12n}{9} = \frac{720 + 0.4n}{15.6}$$

$$10108.8 + 1.872n = 6480 + 3.6n$$

$$3628.8 = 1.728n$$

$$n = 2100$$

From (1),

$$15.6b = 720 + 0.4(2100)$$

$$15.6b = 1560$$

$$b = 100$$

$$\text{Exports of P to ROW} = 20b - 600 - 1200 = 2000 - 1800 = 200$$

$$\text{Exports of X to ROW} = 11b - 4.4b = 6.6b = 660$$

Substituting the values of b and n, we can figure out all the values except for Imports and Exports of ROW to ROW.

The value of exports and imports of ROW to ROW has to be equal.

We calculated the value of n to be 2100.

The exports of ROW to ROW can be calculated as,

$$\text{Exports of ROW to ROW} = 2100 - 840 - 252 = 1008 = \text{Imports of ROW to ROW.}$$

The value of m can be calculated as,

$$m = 200 + 660 + 32 + 1008 = 1900.$$

Filling up the table with all the values, we get,

	P		X		C		ROW		TOTAL	
	IMPORTS	EXPORTS	IMPORTS	EXPORTS	IMPORTS	EXPORTS	IMPORTS	EXPORTS	IMPORTS	EXPORTS
P	0	0	440	600	720	1200	840	200	2000	2000
X	600	440	0	0	48	0	252	660	900	1100
C	1200	720	0	48	0	0	0	32	1200	800
ROW	200	840	660	252	32	0	1008	1008	1900	2100

$$\text{Trade balance of ROW} = \text{Exports} - \text{Imports} = 2100 - 1900 = 200.$$

Hence, the correct answer is option D.

[VIDEO SOLUTION](#)

42. C

In clue 5, we are given that export volumes of P, in IC, to X and C are 600 and 1200, respectively, and P is the only country that exports to C. So, the export from X to C and ROW to C is 0.

In clue 1, we are given that normalised trade balances of P, X and C are 0%, 10%, and -20%, respectively.

For P, since the normalised trade balance is 0% the exports have to be equal to the imports according to the given formula. So, let us assume them to be 10a and 10a.

For X,

$$\frac{E - I}{E + I} \times 100 = 10$$

$$10E - 10I = E + I$$

$$11I = 9E$$

So, let us assume export to be 11b and import to be 9b.

For C,

$$\frac{E - I}{E + I} \times 100 = -20$$

$$5I - 5E = E + I$$

$$2I = 3E$$

So, let us assume export to be 2c and import to be 3c.

	P		X		C		ROW		TOTAL	
	IMPORTS	EXPORTS	IMPORTS	EXPORTS	IMPORTS	EXPORTS	IMPORTS	EXPORTS	IMPORTS	EXPORTS
P	0	0		600		1200			10a	10a
X	600		0	0					9b	11b
C	1200				0	0			3c	2c
ROW										

In clue 2, we are given that 40% of exports of X are to P and 22% of imports of P are from X.

$$40\% \text{ of the exports of X} = \frac{40}{100} \times 11b = 4.4b = \text{exports from X to P}$$

$$22\% \text{ of the imports of P} = \frac{22}{100} \times 10a = 2.2a = \text{imports to P from X}$$

Equating both, we get,

$$4.4b = 2.2a$$

$$a = 2b.$$

We are also given that imports for C are only from P, and we can set the rest of the imports for C to 0.

The total imports for C = $3c = 1200$

$$c = 400$$

This makes the exports for C = $2c = 800$

In clue 3, we are given that 90% of exports of C are to P and 4% are to ROW.

$$90\% \text{ of the exports of C} = \frac{90}{100} \times 2c = 1.8c = 1.8 \times 400 = 720 = \text{exports from C to P}$$

$$4\% \text{ of the exports of C} = \frac{4}{100} \times 2c = 0.08c = 0.08 \times 400 = 32 = \text{exports from C to ROW}$$

$$\text{The exports from C to X} = 800 - 720 - 32 = 48$$

In clue 4, we are given that 12% of exports of ROW are to X, 40% are to P.

If we assume that Imports of ROW are m and exports of ROW are n.

$$12\% \text{ of the exports of ROW} = \frac{12}{100} \times n = 0.12n = \text{exports from ROW to X}$$

$$40\% \text{ of the exports of ROW} = \frac{40}{100} \times n = 0.4n = \text{exports from ROW to P}$$

Placing all the values in the table, we get,

	P		X		C		ROW		TOTAL	
	IMPORTS	EXPORTS	IMPORTS	EXPORTS	IMPORTS	EXPORTS	IMPORTS	EXPORTS	IMPORTS	EXPORTS
P	0	0	4.4b	600	720	1200	0.4n		20b	20b
X	600	4.4b	0	0	48	0	0.12n		9b	11b
C	1200	720	0	48	0	0	0	32	1200	800
ROW		0.4n		0.12n	32	0				

Equating the imports of P and X to the Totals, we get,

For P,

$$4.4b + 720 + 0.4n = 20b$$

$$15.6b = 720 + 0.4n \quad (1)$$

For X,

$$600 + 48 + 0.12n = 9b$$

$$9b = 648 + 0.12n \quad (2)$$

Solving (1) and (2), we get

$$\frac{648 + 0.12n}{9} = \frac{720 + 0.4n}{15.6}$$

$$10108.8 + 1.872n = 6480 + 3.6n$$

$$3628.8 = 1.728n$$

$$n = 2100$$

From (1),

$$15.6b = 720 + 0.4(2100)$$

$$15.6b = 1560$$

$$b = 100$$

$$\text{Exports of P to ROW} = 20b - 600 - 1200 = 2000 - 1800 = 200$$

$$\text{Exports of X to ROW} = 11b - 4.4b = 6.6b = 660$$

Substituting the values of b and n, we can figure out all the values except for Imports and Exports of ROW to ROW.

The value of exports and imports of ROW has to be equal.

We calculated the value of n to be 2100.

The exports of ROW to ROW can be calculated as,

Exports of ROW to ROW = $2100 - 840 - 252 = 1008$ = Imports of ROW to ROW.

The value of m can be calculated as,

$m = 200 + 660 + 32 + 1008 = 1900$.

Filling up the table with all the values, we get,

	P		X		C		ROW		TOTAL	
	IMPORTS	EXPORTS	IMPORTS	EXPORTS	IMPORTS	EXPORTS	IMPORTS	EXPORTS	IMPORTS	EXPORTS
P	0	0	440	600	720	1200	840	200	2000	2000
X	600	440	0	0	48	0	252	660	900	1100
C	1200	720	0	48	0	0	0	32	1200	800
ROW	200	840	660	252	32	0	1008	1008	1900	2100

Total trade of P = $2000 + 2000 = 4000$

Total trade of X = $900 + 1100 = 2000$

Total trade of C = $1200 + 800 = 2000$

So, both countries X and C have the least total trade.

Hence, the correct answer is option C.

[VIDEO SOLUTION](#)

Explanation [43 - 46]:

This is the table given in the question with unknown values of a, b, c, d and e.

Friend	Operator	Outgoing minutes to		Incoming minutes from	
		Operator Xitel	Operator Yocel	Operator Xitel	Operator Yocel
Anu	Xitel	100	a	50	225
Bijay	Xitel	b	200	c	125
Chetan	Yocel	50	175	250	150
Deepak	Yocel	100	150	275	100
Esthan	Yocel	d	100	100	375
Faruq	Yocel	0	e	100	150

We are told that Anu and Bijay use the Xitel operator, and Chetan, Deepak, Esthan and Faruq use the Yocel operator.

The outgoing call from Anu to the Xitel operator has to be the incoming call from Xitel to Bijay because there are only two friends using Xitel, and if one of the Xitel users is calling another Xitel user, then those two have to be Anu and Bijay.

The outgoing to Xitel by Anu = 100

43.50

The incoming from Xitel for Bijay = c = The outgoing to Xitel by Anu = 100

So, the value of c in the above table is 100.

The outgoing to Xitel by Bijay = b

The incoming from Xitel for Anu = 50 = The outgoing to Xitel by Bijay = b

So, the value of b in the above table is 50.

The sum of Outgoing from Anu and Bijay to operator Yocel must be equal to the sum of Incoming minutes from Xitel for Chetan, Deepak, Esthan and Faruq.

The sum of Outgoing from Anu and Bijay to operator Yocel = $a + 200$

The sum of Incoming minutes from Xitel for Chetan, Deepak, Esthan and Faruq = $250 + 275 + 100 + 100 = 725$

Equating them, we get,

$a + 200 = 725$

$a = 525$

So, the value of a in the above table is 525.

The sum of outgoing minutes from Chetan, Deepak, Esthan, and Faruq to operator Xitel must be equal to the sum of Incoming minutes from Yocel for Anu and Bijay.

The sum of outgoing minutes from Chetan, Deepak, Esthan, and Faruq to operator Xitel = $50 + 100 + d + 0 = 150 + d$

The sum of Incoming minutes from Yocel for Anu and Bijay = $225 + 125 = 350$

Equating them, we get,

$150 + d = 350$

$d = 200$

So, the value of d in the above table is 200.

The sum of outgoing minutes from Chetan, Deepak, Esthan, and Faruq to operator Yocel must be equal to the sum of Incoming minutes from Yocel for Chetan, Deepak, Esthan, and Faruq.

The sum of outgoing minutes from Chetan, Deepak, Esthan, and Faruq to operator Yocel = $175 + 150 + 100 + e = 425 + e$

The sum of Incoming minutes from Yocel for Chetan, Deepak, Esthan, and Faruq = $150 + 100 + 375 + 150 = 775$

Equating them, we get,

$$425 + e = 775$$

$$e = 350$$

So, the value of e in the above table is 350.

The final table is,

Friend	Operator	Outgoing minutes to		Incoming minutes from	
		Operator Xitel	Operator Yocel	Operator Xitel	Operator Yocel
Anu	Xitel	100	525	50	225
Bijay	Xitel	50	200	100	125
Chetan	Yocel	50	175	250	150
Deepak	Yocel	100	150	275	100
Esthan	Yocel	200	100	100	375
Faruq	Yocel	0	350	100	150

The duration of calls from Bijay to Ajay is 50 minutes, which is the outgoing minutes to operator Xitel of Bijay.

Hence, the correct answer is 50.

[VIDEO SOLUTION](#)

44. **525**

The incoming from Xitel for Bijay = c = The outgoing to Xitel by Anu = 100

So, the value of c in the above table is 100.

The outgoing to Xitel by Bijay = b

The incoming from Xitel for Anu = 50 = The outgoing to Xitel by Bijay = b

So, the value of b in the above table is 50.

The sum of Outgoing from Anu and Bijay to operator Yocel must be equal to the sum of Incoming minutes from Xitel for Chetan, Deepak, Esthan and Faruq.

The sum of Outgoing from Anu and Bijay to operator Yocel = a + 200

The sum of Incoming minutes from Xitel for Chetan, Deepak, Esthan and Faruq = 250 + 275 + 100 + 100 = 725

Equating them, we get,

$$a + 200 = 725$$

$$a = 525$$

So, the value of a in the above table is 525.

The sum of outgoing minutes from Chetan, Deepak, Esthan, and Faruq to operator Xitel must be equal to the sum of Incoming minutes from Yocel for Anu and Bijay.

The sum of outgoing minutes from Chetan, Deepak, Esthan, and Faruq to operator Xitel = 50 + 100 + d + 0 = 150 + d

The sum of Incoming minutes from Yocel for Anu and Bijay = 225 + 125 = 350

Equating them, we get,

$$150 + d = 350$$

$$d = 200$$

So, the value of d in the above table is 200.

The sum of outgoing minutes from Chetan, Deepak, Esthan, and Faruq to operator Yocel must be equal to the sum of Incoming minutes from Yocel for Chetan, Deepak, Esthan, and Faruq.

The sum of outgoing minutes from Chetan, Deepak, Esthan, and Faruq to operator Yocel = 175 + 150 + 100 + e = 425 + e

The sum of Incoming minutes from Yocel for Chetan, Deepak, Esthan, and Faruq = 150 + 100 + 375 + 150 = 775

Equating them, we get,

$$425 + e = 775$$

$$e = 350$$

So, the value of e in the above table is 350.

The final table is,

Friend	Operator	Outgoing minutes to		Incoming minutes from	
		Operator Xitel	Operator Yocel	Operator Xitel	Operator Yocel
Anu	Xitel	100	525	50	225
Bijay	Xitel	50	200	100	125
Chetan	Yocel	50	175	250	150
Deepak	Yocel	100	150	275	100
Esthan	Yocel	200	100	100	375
Faruq	Yocel	0	350	100	150

The total duration of calls made by Anu to friends having mobile numbers from Operator Yocel is Anu's outgoing minutes to Operator Yocel, which is 525 minutes.

Hence, the correct answer is 525.

45. 350

The incoming from Xitel for Bijay = c = The outgoing to Xitel by Anu = 100

So, the value of c in the above table is 100.

The outgoing to Xitel by Bijay = b

The incoming from Xitel for Anu = 50 = The outgoing to Xitel by Bijay = b

So, the value of b in the above table is 50.

The sum of Outgoing from Anu and Bijay to operator Yocel must be equal to the sum of Incoming minutes from Xitel for Chetan, Deepak, Esthan and Faruq.

The sum of Outgoing from Anu and Bijay to operator Yocel = a + 200

The sum of Incoming minutes from Xitel for Chetan, Deepak, Esthan and Faruq = 250 + 275 + 100 + 100 = 725

Equating them, we get,

$$a + 200 = 725$$

$$a = 525$$

So, the value of a in the above table is 525.

The sum of outgoing minutes from Chetan, Deepak, Esthan, and Faruq to operator Xitel must be equal to the sum of Incoming minutes from Yocel for Anu and Bijay.

The sum of outgoing minutes from Chetan, Deepak, Esthan, and Faruq to operator Xitel = 50 + 100 + d + 0 = 150 + d

The sum of Incoming minutes from Yocel for Anu and Bijay = 225 + 125 = 350

Equating them, we get,

$$150 + d = 350$$

$$d = 200$$

So, the value of d in the above table is 200.

The sum of outgoing minutes from Chetan, Deepak, Esthan, and Faruq to operator Yocel must be equal to the sum of Incoming minutes from Yocel for Chetan, Deepak, Esthan, and Faruq.

The sum of outgoing minutes from Chetan, Deepak, Esthan, and Faruq to operator Yocel = 175 + 150 + 100 + e = 425 + e

The sum of Incoming minutes from Yocel for Chetan, Deepak, Esthan, and Faruq = 150 + 100 + 375 + 150 = 775

Equating them, we get,

$$425 + e = 775$$

$$e = 350$$

So, the value of e in the above table is 350.

The final table is,

Friend	Operator	Outgoing minutes to		Incoming minutes from	
		Operator Xitel	Operator Yocel	Operator Xitel	Operator Yocel
Anu	Xitel	100	525	50	225
Bijay	Xitel	50	200	100	125
Chetan	Yocel	50	175	250	150
Deepak	Yocel	100	150	275	100
Esthan	Yocel	200	100	100	375
Faruq	Yocel	0	350	100	150

The total duration of calls made by Faruq to friends having mobile numbers from Operator Yocel is the outgoing minutes from Faruq to Operator Yocel, which is 350 minutes.

Hence, the correct answer is 350.



46. A

The incoming from Xitel for Bijay = c = The outgoing to Xitel by Anu = 100

So, the value of c in the above table is 100.

The outgoing to Xitel by Bijay = b

The incoming from Xitel for Anu = 50 = The outgoing to Xitel by Bijay = b

So, the value of b in the above table is 50.

The sum of Outgoing from Anu and Bijay to operator Yocel must be equal to the sum of Incoming minutes from Xitel for Chetan, Deepak, Esthan and Faruq.

The sum of Outgoing from Anu and Bijay to operator Yocel = a + 200

The sum of Incoming minutes from Xitel for Chetan, Deepak, Esthan and Faruq = 250 + 275 + 100 + 100 = 725

Equating them, we get,

$$a + 200 = 725$$

$$a = 525$$

So, the value of a in the above table is 525.

The sum of outgoing minutes from Chetan, Deepak, Esthan, and Faruq to operator Xitel must be equal to the sum of Incoming minutes from Yocel for Anu and Bijay.

$$\text{The sum of outgoing minutes from Chetan, Deepak, Esthan, and Faruq to operator Xitel} = 50 + 100 + d + 0 = 150 + d$$

$$\text{The sum of Incoming minutes from Yocel for Anu and Bijay} = 225 + 125 = 350$$

Equating them, we get,

$$150 + d = 350$$

$$d = 200$$

So, the value of d in the above table is 200.

The sum of outgoing minutes from Chetan, Deepak, Esthan, and Faruq to operator Yocel must be equal to the sum of Incoming minutes from Yocel for Chetan, Deepak, Esthan, and Faruq.

$$\text{The sum of outgoing minutes from Chetan, Deepak, Esthan, and Faruq to operator Yocel} = 175 + 150 + 100 + e = 425 + e$$

$$\text{The sum of Incoming minutes from Yocel for Chetan, Deepak, Esthan, and Faruq} = 150 + 100 + 375 + 150 = 775$$

Equating them, we get,

$$425 + e = 775$$

$$e = 350$$

So, the value of e in the above table is 350.

The final table is,

Friend	Operator	Outgoing minutes to		Incoming minutes from	
		Operator Xitel	Operator Yocel	Operator Xitel	Operator Yocel
Anu	Xitel	100	525	50	225
Bijay	Xitel	50	200	100	125
Chetan	Yocel	50	175	250	150
Deepak	Yocel	100	150	275	100
Esthan	Yocel	200	100	100	375
Faruq	Yocel	0	350	100	150

Label the unknown value we need as

x = minutes from Deepak to Chetan

Use the operator groups from the table:

Xitel users: Anu, Bijay.

Yocel users: Chetan, Deepak, Eshan, Faruq.

Chetan's incoming from Yocel = 150 minutes.

Yocel → Chetan can come only from Deepak, Eshan, or Faruq. But the problem states Eshan → Chetan = 0, so

$$(\text{Deepak} \rightarrow \text{Chetan}) + (\text{Faruq} \rightarrow \text{Chetan}) = 150$$

Hence

$$x + (\text{Faruq} \rightarrow \text{Chetan}) = 150 \quad (1)$$

Faruq's outgoing to Yocel = 350 minutes (table). Those 350 minutes are split among Yocel recipients Chetan, Deepak and Eshan. We are told Faruq → Eshan = 200 (given). So

$$(\text{Faruq} \rightarrow \text{Chetan}) + (\text{Faruq} \rightarrow \text{Deepak}) + 200 = 350,$$

hence

$$(\text{Faruq} \rightarrow \text{Chetan}) + (\text{Faruq} \rightarrow \text{Deepak}) = 150 \quad (2)$$

Consider Deepak's incoming from Yocel (table) = 100 minutes. The ones in Yocel that can call Deepak are Chetan, Eshan, and Faruq. But we are told Chetan → Deepak = 0 and Eshan → Deepak = 0, so the only Yocel caller to Deepak is Faruq. Therefore

$$(\text{Faruq} \rightarrow \text{Deepak}) = 100 \quad (3)$$

Substitute (3) into (2):

$$(\text{Faruq} \rightarrow \text{Chetan}) + 100 = 150 \Rightarrow (\text{Faruq} \rightarrow \text{Chetan}) = 50.$$

Now use (1):

$$x + (\text{Faruq} \rightarrow \text{Chetan}) = 150.$$

With Faruq → Chetan = 50, we get,

$$x + 50 = 150$$

$$x = 100$$

Thus Deepak → Chetan = 100 minutes.

Hence, the correct answer is option A.

Quant

47. B

The total number of products is $120 + 135 + 150 + 165 = 570$. Therefore, the total cost must have been $570 \times 240 = 136800$

Let the marked price for the products be X . The January, February, March, and April, selling prices would respectively be $0.8X$, $0.9X$, $0.95X$, and X ; based on the discounts given on each of the months.

The revenue earned from selling the products in each of the months would be: $120 * 0.8X + 135 * 0.9X + 150 * 0.95X + 165 * X = 525X$.

The profit earned would be: Revenue - Cost, and therefore,

$$525X - 136800 = 138825$$

$$525X = 275625$$

$$X = \frac{275625}{525} = 525$$

The correct answer is option B, Rs. 525.

48. B

Let the per hour productivity of each member from teams A, B, and C, be a , b , and c units respectively.

Let W be the total units of work that each of the teams does. We have:

$$W = 40 * 5 * a = 50 * 8 * b = 4 * 10 * c$$

$$\text{This gives, } a = \frac{W}{200}, b = \frac{W}{400}, \text{ and } c = \frac{W}{40}$$

When two members from team A, three members from team B, and one member from team C together start the job, the total per hour productivity for the first 23 hours will be;

$$\frac{2W}{200} + \frac{3W}{400} + \frac{W}{40} = \frac{17W}{400}$$

$$\text{In 23 hours, the work done will be } \frac{23 * 17W}{400} = \frac{391W}{400}$$

$$\text{The remaining work, after 23 hours, is } W - \frac{391W}{400} = \frac{9W}{400}$$

$$\text{The current per hour efficiency of the group that comprises members from only teams A and B, is } \frac{2W}{200} + \frac{3W}{400} = \frac{7W}{400}$$

$$\text{In the next hour, the members from A and B will finish } \frac{7W}{400} \text{ units of work and } \frac{9W}{400} - \frac{7W}{400} = \frac{2W}{400} \text{ units of work would remain.}$$

$$\text{This work has to be finished by newly added members from team B in one hour, therefore, the number of new members required from team B would be } \frac{2W}{400} \div \frac{W}{400} = 2$$

Option B is the correct answer.

49. 700

Let the total number of students who chose science, arts, and commerce streams be S , A , and C , respectively.

We have,

$$S + A + C = 1500, \text{ such that, } C = 1500 - S - A \text{(1)}$$

Also, $1100S + 1000A + 800C = 15,50,000$, which can be simplified by dividing by 100;

$$11S + 10A + 8C = 15500 \text{(2)}$$

Substituting the value of C from equation (1) in equation (2), we have

$$11S + 10A + 8(1500 - S - A) = 15500$$

$$\Rightarrow 3S + 2A = 3500$$

Since $S \leq A$, the maximum value of S (if possible), would occur when $S = A$, therefore,

$$5S = 3500, \text{ or } S = 700.$$

50. 65

Let the first term of the arithmetic progression be a , and the common difference be d .

We have:

$$(a + 3d) + (a + 6d) + (a + 9d) = 99$$

$$\Rightarrow 3a + 18d = 99 \text{ or } a + 6d = 33 \text{(1)}$$

We are also told that the sum of the first fourteen terms is 497,

$$\frac{14}{2}(2a + (14 - 1)d) = 497$$

$$\Rightarrow (2a + 13d) = \frac{497}{7}$$

$$\Rightarrow 2a + 13d = 71 \dots(2)$$

Solving equations (1) and (2), we get;

$$a = 3 \text{ and } d = 5$$

The first five terms would therefore be: 3, 8, 13, 18, 23, and their sum would be $3 + 8 + 13 + 18 + 23 = 65$.

[VIDEO SOLUTION](#)

100+ CAT Quant Questions



51.444

Let the walking speed of Ankita be $5x$, implying that her running speed (which is 40% more than her walking speed), is $1.4 \times 5x = 7x$.

Thus, the ratio of her walking and running speeds is $5 : 7$. Therefore, the ratio of the time taken by Ankita to cover a fixed distance walking and running would be $7 : 5$.

She takes 3 hours 30 minutes, or 3.5 hours to walk from B to C. Since Ankita is running from B to C in the second scenario, her time will reduce inversely to the ratio of her speed. The time taken by her to run from B to C would be $\frac{3.5}{7} \times 5 = 2.5$ hours.

She takes 3 hours 30 minutes, or 3.5 hours to run from A to B. Since Ankita is walking from A to B in the second scenario, her time will increase inversely to the ratio of her speed. The time taken by her to walk from A to B would be $\frac{3.5}{5} \times 7 = 4.9$ hours.

Therefore, the total time it takes for Ankita in the second scenario is $4.9 + 2.5 = 7.4$ hours. This gives $7.4 \times 60 = 444$ minutes.

[VIDEO SOLUTION](#)

52.A

Let the thousands, hundreds, tens, and units digits of the number be a, b, c , and d , respectively.

The sum of the digits in the thousands, hundreds, and tens places is 15; $a + b + c = 15 \dots(1)$

The sum of the digits in the hundreds, tens, and units places is 16; $b + c + d = 16 \dots(2)$

Solving equations (1) and (2), we get $d = a + 1$.

The digit in the tens place is 6 more than the digit in the units place; $c = d + 6 = a + 1 + 6 = a + 7 \dots(3)$

Substituting the value of c in terms of a in equation (1), we get; $a + b + a + 7 = 15$ or $b = 8 - 2a$

Thus, the number is: $\underline{a} \underline{8 - 2a} \underline{a + 7} \underline{a + 1}$

We know that since the number is a four digit number, $a \geq 1$. But all the four digits have to be less than or equal to 9 but greater than or equal to 0, thus,

$8 - 2a \geq 0$ gives $a \leq 4$ and $a + 7 \leq 9$ gives $a \leq 2$.

Thus, a can be 1 or 2.

The two possible values for the number, therefore, are, 1682 and 2493, when we put $a = 1$ and $a = 2$, respectively.

The difference between the only two possible values is, $2493 - 1682 = 811$

Option A is the correct answer.

[VIDEO SOLUTION](#)

53.B

Let the initial speed of Rahul be x kilometres per hour. Since he travels the distance usually in 6 hours, (5 pm to 11 pm), the total distance should be $6 \times x = 6x$ kilometres.

Let the distance after which Rahul stops for some duration in both scenarios be y kilometres.

In the first scenario, he stops for 20 minutes or $\frac{1}{3}$ hours, therefore, the travel time would be $6 - \frac{1}{3} = \frac{17}{3}$ hours. We have,

$$\frac{k}{x} + \frac{6x - k}{x + 3} = \frac{17}{3}$$

$$\Rightarrow \frac{6x^2 + 3k}{x^2 + 3x} = \frac{17}{3}$$

$$\Rightarrow 18x^2 + 9k = 17x^2 + 51x$$

$$\Rightarrow x^2 = 51x - 9k \dots(1)$$

Similarly, in the second scenario, he stops for $20 + 10 = 30$ minutes or $\frac{1}{2}$ hours, therefore, the travel time would be $6 - \frac{1}{2} = \frac{11}{2}$ hours. We have,

$$\frac{k}{x} + \frac{6x - k}{x + 5} = \frac{11}{2}$$

$$\Rightarrow \frac{6x^2 + 5k}{x^2 + 5x} = \frac{11}{2}$$

$$\Rightarrow 12x^2 + 10k = 11x^2 + 55x$$

$$\Rightarrow x^2 = 55x - 10k \dots (2)$$

From equations (1) and (2), we have;

$$55x - 10k = 51x - 9k \text{ or}$$

$$4x = k.$$

Substituting the value of k in equation (1), we have

$$x^2 = 51x - 36x \text{ or, since } x \text{ is positive, } x = 15. \text{ Therefore, option B is the correct answer.}$$

 VIDEO SOLUTION

54. A

Let the rate of flow of water from pipe A be $4x$ units per hour. From the ratio of the rates of flow of water (4:9:36), we have that the rate of flow of water from pipe B should be $9x$ units per hour, and from pipe C should be $36x$ units per hour.

Since pipe A fills the empty tank in 15 hours, the capacity of the tank must be $15 \times 4x = 60x$ units.

When all three pipes work together, their combined capacity would be $4x + 9x + 36x = 49x$ units per hour.

Therefore, the time it would take for the three pipes to fill the empty tank working together is: $\frac{60x}{49x}$ hours

In minutes, this time is equivalent to $\frac{60x}{49x} \times 60 \approx 73.47$ minutes.

Option A is the closest, and is the correct answer.

 VIDEO SOLUTION

55. D

Let $(x^2 + 3x)$ be equal to k . We have,

$$f(x) = k(k + 2) = k^2 + 2k$$

$$\text{Therefore, } \sqrt{f(x) + 1} = \sqrt{k^2 + 2k + 1} = \sqrt{(k + 1)^2} = k + 1 = 9701$$

We get $k = 9700$.

$$\text{Thus, } x^2 + 3x = 9700 \text{ or } x^2 + 3x - 9700 = 0$$

Since x is real, the discriminant of the above quadratic has to be greater than or equal to zero.

We find that $3^2 + 4 * 9700 \geq 0$ and therefore the quadratic has real roots.

$$\text{The sum of the roots will be } -\frac{b}{a} = -\frac{3}{1} = -3$$

Option D is the correct answer.

 VIDEO SOLUTION

100+ CAT DILR Questions



56. C

$$\text{Let } y = \frac{2x - 3}{2x^2 + 4x - 6}$$

$$\Rightarrow 2yx^2 + 4yx - 6y = 2x - 3$$

$$\Rightarrow 2yx^2 + (4y - 2)x - 6y + 3 = 0$$

Equation (1) is a quadratic in x , where x is real. Therefore, the discriminant of the quadratic has to be greater than or equal to zero.

$$(4y - 2)^2 + 4 \times 2y \times (6y - 3) \geq 0$$

$$16y^2 + 4 - 16y - 24y + 48y^2 \geq 0$$

$$64y^2 - 40y + 4 \geq 0$$

$$16y^2 - 10y + 1 \geq 0$$

$$\text{The roots of the quadratic above will be } \frac{10 \pm 6}{32} = \frac{1}{2} \text{ or } \frac{1}{8}$$

Since the coefficient of y^2 is positive, the quadratic will be less than zero in the range $\left(\frac{1}{8}, \frac{1}{2}\right)$

The quadratic will be greater than or equal to zero otherwise.

Therefore, the domain the quadratic, or possible values of y , which is the range of $f(x)$, will be,

$$\left(-\infty, \frac{1}{8}\right] \cup \left[\frac{1}{2}, \infty\right)$$

Option C is the correct answer.

VIDEO SOLUTION

57. B

$(10^{50} + 10^{25} - 123)$ can be written as $10^{50} + (10^{25} - 123)$

$1000 - 123$ gives 877.

$10000 - 123$ gives 9877.

$100000 - 123$ gives 99877.

$1000000 - 123$ gives 999877.

And so on.

Considering $10^n - 123$, which has a total of n digits; in general, of the n total digits; the digit 7 is repeating 2 times, digit 8 is repeating 1 time, and digit 9 is repeating $(n - 3)$ times.

Also, when added to 10^{50} , the cumulative increase in the sum of the digits of the resultant number would be 1, which would come from the leftmost digit of the result. All other newly added digits will be zero.

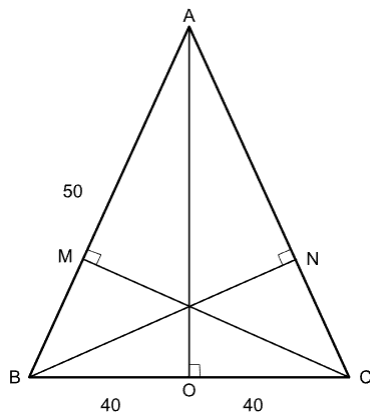
Thus, the sum of the digits of the result will be given by;

$$[7 * 2] + [1 * 8] + [(25 - 3) * 9] + [25 * 0] + [1 * 1] = 14 + 8 + 198 + 1 = 221$$

The correct answer is option B.

VIDEO SOLUTION

58. 126



The altitudes CM and BN will be equal as the triangle $\triangle ABC$ is isosceles.

Further, in $\triangle AOB$, using Pythagoras Theorem, we have

$$AO^2 + OB^2 = AB^2, \text{ therefore, } AO^2 = 50^2 - 40^2$$

Thus, $AO^2 = 900$ or $AO = 30$ cm.

We know that the ratio of altitudes in a triangle is the inverse of the ratio of sides; therefore, the ratio of altitudes of the triangle will be;

$$\frac{1}{50} : \frac{1}{50} : \frac{1}{80} = 8 : 8 : 5$$

Therefore, $CM = BN = 8x$ and $AO = 5x$.

Since $5x = 30$, we have $x = 6$, and the sum of altitudes, in centimetres, will be $8x + 8x + 5x = 21x = 21 * 6 = 126$.

VIDEO SOLUTION

59. A

$$\left(x + \frac{1}{x}\right)^2 = x^2 + \frac{1}{x^2} + 2 = 25 + 2 = 27$$

$$\text{Therefore, } \left(x + \frac{1}{x}\right) = \sqrt{27} = 3\sqrt{3}$$

$$\text{Also, } \left(x + \frac{1}{x}\right)^3 = x^3 + \frac{1}{x^3} + 3\left(x + \frac{1}{x}\right)$$

$$\text{Therefore, } x^3 + \frac{1}{x^3} = (3\sqrt{3})^3 - 9\sqrt{3} = 72\sqrt{3}$$

$$\text{Also, } \left(x^2 + \frac{1}{x^2}\right)^2 = x^4 + \frac{1}{x^4} + 2$$

$$\text{Therefore, } x^4 + \frac{1}{x^4} = (25)^2 - 2 = 623$$

$$\text{Lastly, } \left(x^4 + \frac{1}{x^4}\right) \left(x^3 + \frac{1}{x^3}\right) = x^7 + \frac{1}{x^7} + x + \frac{1}{x}$$

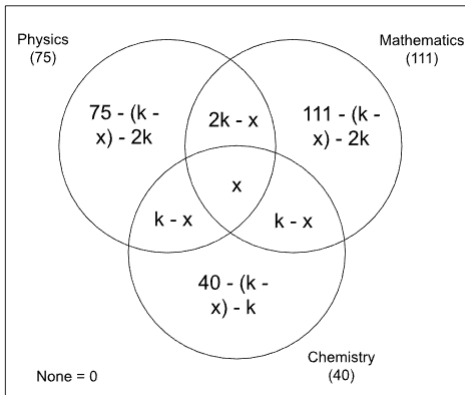
$$\text{Therefore, } x^7 + \frac{1}{x^7} = 623 * (72\sqrt{3}) - 3\sqrt{3} = 44853\sqrt{3}$$

Option A is the correct answer.

VIDEO SOLUTION

60. B

Based on the information provided, we can make the following diagram;



We know that the sum of all entries in the Venn diagram should be 150. Therefore,

$$111 + 75 - 2k + 40 - (k - x) - k + 0 = 150$$

$$76 = 4k - x, \text{ where } x \geq 1$$

We must maximise the number of students who chose Physics but not Mathematics. This number is equal to $75 - 2k$. To maximise this we must minimise k .

$$\text{We have } 4k = 76 + x \text{ or } k = 19 + \frac{x}{4}$$

Since x and k are both integers, x should be divisible by 4. To minimise k , we will take the minimum value of x , which is 4.

$$\text{Therefore, } k = 19 + \frac{4}{4} = 20.$$

Thus, the maximum possible number of students who chose physics but not mathematics, is $75 - 2 * 20 = 35$.

VIDEO SOLUTION

100+ CAT VARC Questions



61. D

Since the base of the logarithm has to be greater than zero and cannot be 1, x has to be greater than 3 and cannot be 4. Also, since $x^2 - 9 > 0$, x will be greater than 3.

The equation can be rewritten as;

$$\log_{x-3}(x^2 - 9) - \log_{x-3}(x + 1) = 2$$

Or,

$$\log_{x-3} \frac{x^2 - 9}{x + 1} = 2$$

$$\Rightarrow \frac{(x + 3)(x - 3)}{x + 1} = (x - 3)^2$$

$$\Rightarrow \frac{(x + 3)}{x + 1} = (x - 3)$$

$$\Rightarrow (x + 3) = (x - 3)(x + 1)$$

$$\Rightarrow x + 3 = x^2 - 3x + x - 3$$

$$\Rightarrow x^2 - 3x - 6 = 0$$

$$\text{The roots of the quadratic above are } \frac{3 \pm \sqrt{3^2 + 24}}{2}$$

$$\text{The negative value of } x \text{ will not be possible, the positive value that satisfies is } \frac{3 + \sqrt{33}}{2}$$

The correct answer is option D.

VIDEO SOLUTION

62. C

Let the average salary of managers be x and let the average salary of engineers be y . The total salary of all the employees will be $(5 + 25) * 60000 = 1800000$

$$\text{We have, } 5x + 25y = 1800000 \dots (1)$$

If the average salary of all the employees increases by 5%, the total salary of all the employees will also increase by 5%, because the total number of employees remains the same. The new total salary will be, $1800000 \times 1.05 = 1890000$

Also, the average salary of all the managers has increased by 20% and has become $1.2x$, we have

$$5 * 1.2x + 25y = 1890000, \text{ or } 6x + 25y = 1890000 \dots (2)$$

Subtracting equation (1) from equation (2), we get,

$$x = 90000$$

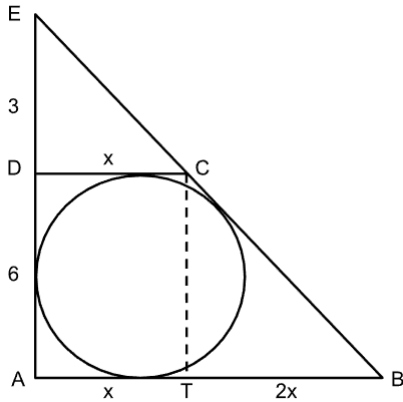
$$\text{Which gives } 25y = 1800000 - 90000 * 5 \text{ or } y = \frac{1350000}{25} = 54000$$

Therefore, the correct answer is option C.

[VIDEO SOLUTION](#)

63. A

We extend BC and AD to meet at E , and based on the information provided, can construct the diagram as follows:



$DC = AT = x$ and $BT = 2x$ such that $AB = 3DC$. $\angle A = 90^\circ$ and triangles ABE and DCE are similar.

Since AD is equal to the diameter of the circle, and $AE = 3DE$, we get $DE = 3$ cm.

In right-angled triangle DCE , we have $CE = \sqrt{x^2 + 9}$

In right-angled triangle BCT , we have $BC = \sqrt{4x^2 + 36} = 2\sqrt{x^2 + 9}$

Since $BE = BC + CE$, we have $BE = 3\sqrt{x^2 + 9}$

The radius of the incircle of a right-angled triangle is given by $\frac{s_1 + s_2 - \text{hypotenuse}}{2}$

$$\text{Therefore, } 3 = \frac{3x + 9 - 3\sqrt{x^2 + 9}}{2}$$

Which gives,

$$-x - 1 = -\sqrt{x^2 + 9} \text{ or } x^2 + 2x + 1 = x^2 + 9 \text{ or } x = 4.$$

Lastly, the area of the trapezium would be $\frac{1}{2} \times (x + 3x) \times 6 = 12x = 12 \times 4 = 48$ square centimetres.

[VIDEO SOLUTION](#)

64. 112

$$12^{12x} \times 4^{24x+12} \times 5^{2y} = 8^{4z} \times 20^{12x} \times 243^{3x-6}$$

On rewriting after prime factorisation, we get,

$$2^{24x} \times 3^{12x} \times 2^{48x+24} \times 5^{2y} = 2^{12z} \times 2^{24x} \times 5^{12x} \times 3^{15x-30}$$

Since LHS = RHS, the corresponding powers must be equal. We have,

Powers of 3: $12x = 15x - 30$ or $x = 10$, and

Powers of 2: $48x + 24 = 12z$ or $4x + 2 = z$ or $z = 42$

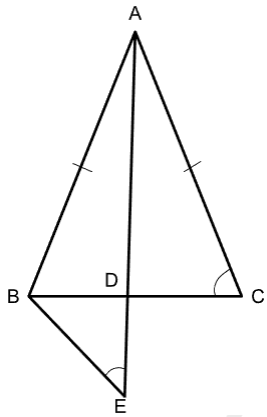
Powers of 5: $2y = 12x$ or $y = 6x$ or $y = 60$

Therefore, $x + y + z = 10 + 42 + 60 = 112$

[VIDEO SOLUTION](#)

65. C

Based on the information provided, we can construct the following diagram;



We have $AB = AC = 12$ cm, and $AD = 8$ cm. And $\angle ACB = \angle AEB$

Since the triangle is isosceles, $\angle ACB = \angle AEB = \angle ABC$ (or $\angle ABD$)

In $\triangle ABE$ and $\triangle ADB$,

$\angle BAE = \angle DAB$ and $\angle AEB = \angle ABD$. Therefore, by AA similarity criterion, $\triangle ABE \sim \triangle ADB$

Hence, we get,

$$\frac{AB}{AD} = \frac{AE}{AB} \text{ or } \frac{12}{8} = \frac{AE}{12} \text{ or } AE = 18 \text{ cm.}$$

[VIDEO SOLUTION](#)

CAT Expected Questions PDF



66.16

Vessel A contains 60 litres of pure alcohol, and Vessel B contains 60 litres of pure water. Let the amount taken out from vessel A be x litres. The contents in Vessel A now are $60 - x$ litres of alcohol, and in Vessel B now are 60 litres of water and x litres of alcohol.

After mixing, when the same quantity x is taken out from Vessel B, the alcohol that would be taken out would be $\left(\frac{x}{60+x}\right)^{\text{th}}$ of x , which is $\frac{x^2}{60+x}$ litres.

The initial total quantity of 60 litres has been restored in A after replacement. The total amount of alcohol in Vessel A after the process is complete is:

$$60 - x + \frac{x^2}{60+x} = \frac{(60+x)(60-x) + x^2}{60+x} = \frac{3600}{60+x}$$

The total quantity in vessel A is 60 litres where the ratio of alcohol to total quantity is $\frac{15}{15+4} = \frac{15}{19}$

Therefore,

$$\frac{3600 \div (60+x)}{60} = \frac{15}{19}$$

$$\Rightarrow \frac{60}{60+x} = \frac{15}{19}$$

$$\Rightarrow x = 16$$

Thus, the quantity taken out and replaced in the two iterations is 16 litres.

[VIDEO SOLUTION](#)

67.272

Let the initial total number of coins in box A be $17x$, since the ratio of the number of coins in boxes A and B is $17 : 7$, the initial total number of coins in box B would be $7x$.

After 108 coins are shifted from box A to box B, the number of coins in box A would be $17x - 108$ and in box B would be $7x + 108$. We have,

$$\frac{17x - 108}{7x + 108} = \frac{37}{20}$$

$$\Rightarrow 340x - 2160 = 259x + 3996$$

$$\Rightarrow 81x = 6156 \text{ or } x = 76$$

Therefore, the total number of coins in the two boxes combined are $(17 + 7) * 76 = 1824$, half of which is 912. The number of coins in box B currently is $7 * 76 + 108 = 640$. The number of coins needed to make the coins equal in both the boxes is, $912 - 640 = 272$.

Thus, 272 coins will be transferred from box A to box B to make the number of coins equal (1:1) in both the boxes.

[VIDEO SOLUTION](#)

68.397

We have $p + q + r = 900$, and r is a perfect square lying between 150 and 500.

Let q be a non-variable number that does not have a range.

We have $0.3q \leq p \leq 0.7q$, or $q + 0.3q \leq p + q \leq q + 0.7q$ or $1.3q \leq p + q \leq 1.7q$

Also, we can rewrite $p + q = 900 - r$

Therefore, $1.3q \leq 900 - r \leq 1.7q$

Since the extreme values (maximum and minimum) of p depend on the value of q ($0.3q$ and $0.7q$), we can maximise or minimise r to find the extreme values of p .

Since $1.3q \leq 900 - r$, the minimum possible value of q would be when $1.3q = 900 - r$ and when r is maximum. The maximum value of r is 484. We get, $1.3q = 900 - 484$, or $q = 320$.

This gives the minimum possible value of p as $0.3 \times 320 = 96$.

Since $900 - r \leq 1.7q$, the maximum possible value of q would be when $900 - r = 1.7q$ and when r is minimum. The minimum value of r is 169. We get $1.7q = 900 - 169$ or $q = 430$

This gives the maximum possible value of p as $0.7 \times 430 = 301$

The sum of the maximum and minimum values of p is $96 + 301 = 397$.

 VIDEO SOLUTION